

# The Convex Origin of Fixed Costs<sup>†</sup>

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## Abstract

Are models with fixed and convex adjustment costs different? I show they are Fenchel-dual: a distribution  $G$  of fixed costs generates a convex cost  $C$  whose curvature reflects selection over heterogeneous adjusters; the deterministic fixed cost is the knife-edge limit, convexity is generic. The duality delivers value-function differentiability and a smooth, closed-form first-order condition  $S = C'(\varphi)$  in place of threshold machinery; it microfounds the adjustment-hazard exponent as  $\nu = 2\gamma$  (profit curvature  $\times$  cost-concentration) and decomposes the Phillips curve slope as  $\kappa = (1 + 2\gamma)\bar{\Lambda}$ . The construction unifies menu costs, lumpy investment, rational inattention, discrete choice, and control costs as images of one  $G$ , and extends to general-equilibrium frictions. In the pricing application, taking [Alvarez, Lippi, and Oskolkov \(2022\)](#)'s benchmark of  $\nu \approx 2$  from price-change microdata pins down  $\hat{\gamma} = 1$ , so that selection accounts for roughly two-thirds of the Phillips curve slope — emerging as the dominant channel of monetary non-neutrality.

**Keywords:** Fixed costs, convex adjustment costs, Fenchel duality, menu costs, discrete choice, rational inattention, Phillips curve.

**JEL codes:** D81, E31, E22, E32.

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# 1 Introduction

A central puzzle in macroeconomics is the gap between micro behaviour and aggregate dynamics. At the individual level, adjustment is often lumpy and discontinuous: firms change prices in sudden jumps, invest in infrequent spikes, and workers transition between labour-force states in discrete moves. Yet aggregate dynamics are smooth and gradual — inflation responds continuously to shocks, capital accumulates in humps, and labour markets absorb disruptions fluidly. How micro lumpiness aggregates into macro smoothness — and whether the micro structure still matters for aggregate conclusions — has shaped the literature for four decades. Two modelling traditions capture the tension. *Fixed* adjustment costs reproduce the micro jumps: agents stay inactive until a threshold is crossed, then act. *Convex* adjustment costs reproduce the aggregate smoothness: actions respond gradually and probability-weighted to the state. The two deliver sharply different predictions for monetary non-neutrality, the Phillips curve slope, and the welfare cost of adjustment frictions.

Are models with fixed and convex adjustment costs different in nature? This paper shows they are *ex-ante equivalent*: before the cost realisation, the two formulations are interchangeable. The key insight is that when fixed costs are heterogeneous — drawn from a distribution  $G$  — the threshold rule under  $G$  and a smooth choice of  $\varphi$  paid for by a convex cost  $C(\varphi)$  deliver the same expected payoff and the same adjustment probability. The two formulations are Fenchel duals:  $C$  is the quantile integral of  $G$ , and the payoff is its conjugate over the surplus. The realised action is binary in both formulations; what coincides is the smooth ex-ante extensive margin. The mechanism is selection: agents with low cost draws adjust first, so the marginal non-adjuster faces a higher cost draw than the last. This increasing marginal cost is convexity.

The duality has structural implications at both the firm and aggregate level. The deterministic fixed cost — where every firm faces the same cost — is a knife-edge singularity; any small amount of cost heterogeneity, however slight, makes the induced cost over the adjustment probability strictly convex (Proposition 2). At the aggregate level, the shape of  $G$  is the relevant dial: uniform  $G$  gives textbook quadratic probability adjustment costs (Khan and Thomas, 2008); concentrated  $G$  gives the  $S$ s rule of Golosov and Lucas (2007); dispersed  $G$  approximates Calvo (1983) in the frequency-calibrated limit of Proposition 9. These are not different kinds of models but different shapes of one distribution. This also provides a partial-equilibrium foundation for the irrelevance result of Khan and Thomas (2008): with uniform fixed costs, each firm’s ex-ante extensive margin is already convex, so aggregation is not the source of the smoothness.

The contribution is not the duality itself — the binary-choice version of the Williams-

Daly-Zachary identity (McFadden, 1978, 1981) is well known in discrete choice, and the convex-cost-over-adjustment-probability object in the menu-cost case coincides with the §2.2 cost of Alvarez, Lippi, and Oskolkov (2022). The contribution lies in what the duality makes possible. The closed-form construction  $G \mapsto C$  is environment-free and ports across applications (Theorem 1). The smoothness of  $C$  yields a transparent route to value-function differentiability (Theorem 2), with a closed-form smooth first-order condition  $S = C'(\varphi)$  in place of threshold-specific machinery. The  $C^1$  regularity itself is standard, following from general dynamic-programming arguments (Santos 1991) and obtained in lumpy settings via smooth-pasting (Baley and Blanco 2021); the cost object coincides with that of Alvarez, Lippi, and Oskolkov (2022), recovered there from the hazard. The contribution is the route: the map  $G \mapsto C$  is environment-free, and it yields the closed-form FOC  $S = C'(\varphi)$  in any environment admitting a scalar adjustment surplus, putting the dynamic model directly into FOC-based heterogeneous-agent solvers (Reiter, 2009; Auclert, Rognlie, and Straub, 2020) without threshold-specific machinery. The same smoothness is the operational input to the central application: a fully global, non-linear solution of the entire Calvo–Goloso–Lucas spectrum by the repeated transition method (RTM) of Lee (2026) (Section 4.5), which linearised methods cannot deliver, producing state-dependent monetary transmission conditioned on the cross-sectional dispersion of price gaps. Under the fixed-cost primitive,  $C(\varphi^*)$  acquires a concrete resource-cost reading — the expected physical cost paid by adjusters — giving the abstract information penalty of rational inattention an estimable physical counterpart.

The reverse direction of the duality is informative for researchers who take a convex penalty  $C$  as the primitive — the rational inattention and control cost traditions: it reveals the cost function as an implicit assumption on  $G$  rather than a free modelling choice (Proposition 4). The quadratic adjustment cost in Khan and Thomas (2008) and the quadratic attention penalty in Gabaix (2014) are what uniform  $G$  over  $[0, \bar{\xi}]$  induces; the binary rational inattention cost of Matějka and McKay (2015) (a generalised Kullback-Leibler divergence from the no-adjustment prior) is what exponential  $G$  induces, with the attention shadow price  $\lambda$  identified as the rate parameter of the cost distribution; power law  $G$  delivers a Tsallis  $q$ -divergence (Tsallis, 1988). In the multi-action extension, i.i.d. Gumbel payoff shocks recover the multinomial logit of McFadden (1974) ( $\xi$  reinterpreted as a real-valued payoff shock rather than a non-negative cost). Five literatures — menu costs, lumpy investment, discrete choice, rational inattention, and control costs (Mattsson and Weibull, 2002; Hofbauer and Sandholm, 2002) — are not merely connected: the cost specification each uses is the parametric image of an implicit assumption on  $G$ , an assumption that becomes explicit, testable, and in principle recoverable from cost-distribution data (as Oskolkov and Lippi

2026 do for plant-level investment).

The pricing application is developed in greatest depth. Under the standard quadratic approximation to the repricing surplus,  $S(x) = Bx^2$  in the price gap  $x$ , the duality delivers closed-form aggregation. A power law family  $G(\xi) = (\xi/\bar{\xi})^\gamma$  parametrises a continuous spectrum between Calvo (dispersed costs, weak selection) and Golosov-Lucas (concentrated costs, strong selection); the slope of the aggregate price response to a nominal shock,  $\kappa$ , decomposes (Proposition 10) into *frequency* (how many firms reprice) and *selection* (how well repricing is targeted), with  $\gamma$  governing the balance. Alvarez, Le Bihan, and Lippi (2016) show that frequency and kurtosis of price changes are sufficient statistics for non-neutrality within their model class; the present framework provides the structural layer from which those observable moments are derived, since  $G$  determines the repricing hazard and hence both moments. Solving the full dynamic model globally across the spectrum (Section 4.5) yields the central economic finding: monetary transmission is state-dependent in the cross-sectional dispersion of price gaps. In regimes of high dispersion — more firms sitting near the adjustment boundary — both output and inflation respond more strongly to demand shocks, with the selection margin amplifying the impact; this is a margin of behaviour that linearised methods eliminate by construction. The global solution also verifies that the analytical decomposition holds as a tight numerical approximation in the dynamic model, with the residual error governed by the deviation of the dynamic surplus from exact quadraticity. The benchmark calibration anchors  $\gamma$  to the heterogeneity-corrected hazard exponent of Alvarez, Lippi, and Oskolkov (2022) via the structural mapping  $\nu = 2\gamma$  of the Remark after Theorem 1: their recovered  $\nu \approx 2$  from price-change microdata implies  $\hat{\gamma} = 1$ , with  $\kappa = 3\bar{\Lambda}$  and a selection share of about 67% at the calibrated repricing frequency. Selection is the dominant component of monetary non-neutrality, with frequency accounting for the remaining one-third.

The duality applies more broadly to any extensive-margin decision with heterogeneous costs — firm entry, search and matching, labour-force participation, technology adoption, and credit default — discussed in Section 5.

**Related literature** The analysis connects several literatures. On macroeconomic frictions and pricing: Caplin and Spulber (1987); Dotsey, King, and Wolman (1999); Golosov and Lucas (2007); Midrigan (2011); Nakamura and Steinsson (2010); Alvarez, Le Bihan, and Lippi (2016); Vavra (2014); on lumpy investment, Thomas (2002); Gourio and Kashyap (2007); Oskolkov and Lippi (2026). The Vavra (2014) sufficient statistic is endogenous in  $G$ , which governs both the cross-sectional variance and the friction generating it. The repricing hazard  $\Lambda(x)$  realises the “adjustment function” primitive of Caballero and Engel (1993, 1999): its

shape is not assumed but derived from  $G$ , micro-founding the adjustment-function approach. Caballero and Engel (2007) show that the variance of idiosyncratic shocks relative to the inaction band governs amplification; Proposition 12 states an analogous monotonicity in  $\gamma$ . Most closely related, Alvarez, Lippi, and Oskolkov (2022) establish in a continuous-time sticky-price setting that a generalised hazard function  $\Lambda(x)$  admits two equivalent rationalisations — a random menu cost distribution  $G$  and a convex flow cost  $c(\ell)$  on Poisson intensity — connected through their common implementation of  $\Lambda$ . Theorem 1 below builds on and complements this result by establishing the duality directly between  $G$  and  $C$  in discrete time, with an explicit closed-form construction and applicability to any extensive-margin decision (see the remark after Theorem 1).

The Williams-Daly-Zachary theorem (McFadden, 1974, 1978, 1981) establishes the same algebraic skeleton in discrete choice; the contributions here — the  $G$ -indexed taxonomy, the Khan-Thomas foundation, the pricing spectrum — are absent there. The fixed cost model is a control cost model (Mattsson and Weibull, 2002; Hofbauer and Sandholm, 2002; Gabaix, 2014) with penalty endogenously determined by  $G$ ; binary rational inattention (Sims, 2003; Matějka and McKay, 2015) with a degenerate prior on inaction yields KL, recovered here from exponential  $G$ , and the multinomial logit’s negative Shannon entropy from i.i.d. Gumbel costs. Concurrent independent work by Bloedel, Denti, and Pomatto (2025) uses Fenchel duality to analyse a rational inattention model with a different primitive — an information-acquisition cost rather than a distribution of physical fixed costs.

**Roadmap** Section 2 presents the duality theorem, characterises the induced cost function, establishes universality, develops the entropy correspondence, and applies the result to investment. Section 3 develops the pricing spectrum. Section 4 calibrates  $\gamma$  from micro price moments, quantifies the frequency-selection decomposition, and reports the global non-linear solution of the dynamic model (Section 4.5) that the differentiability theorem makes feasible — verifying the analytical decomposition as a tight approximation and producing the state-dependent transmission finding. Section 5 discusses further applications, interpretation, and scope. Section 6 concludes.

## 2 The duality

### 2.1 Primitives

Consider an agent who faces a binary decision: take an action (“adjust”) or maintain the status quo (“do nothing”). Adjusting yields a gross surplus  $S \geq 0$  but requires paying a

fixed cost  $w\xi$ , where  $w > 0$  is a wage or unit cost and  $\xi$  is a random draw from a distribution  $G$  supported on  $[0, \bar{\xi}]$  with  $\bar{\xi} \in (0, \infty]$  (the case  $\bar{\xi} = \infty$  covers unbounded supports such as the exponential), with density  $g > 0$  on its support and finite mean  $E_G[\xi] < \infty$ . All integrals against  $dG$  are taken over the support of  $G$ . Not adjusting yields zero. The randomness in  $\xi$  captures heterogeneity in the cost of adjustment. This heterogeneity may be across agents (a cross-section of firms facing different menu costs) or across time (a single firm drawing a new cost each period, as in [Dotsey, King, and Wolman 1999](#)).

This setup is minimal. It strips away the details of particular applications to isolate the structure common to all of them. In the investment application,  $S$  is the surplus from investing at the optimal scale. In the pricing application of Section 3,  $S$  is the gain from closing the price gap. Because the setup is general, the duality theorem applies to any extensive-margin decision where the agent's cost of acting is uncertain.

**Definition 1** (Fixed cost formulation).

*Given the payoff  $S$ , the following defines the fixed cost payoff  $\Pi^{FC}(S)$  and the threshold rule  $\xi^*$ : The agent observes the realised cost  $\xi$  and adjusts if and only if  $w\xi \leq S$ :*

$$\Pi^{FC}(S) = \int_0^{\bar{\xi}} \max\{S - w\xi, 0\} dG(\xi) = \int_0^{\min\{S/w, \bar{\xi}\}} (S - w\xi) dG(\xi). \quad (1)$$

*The agent adjusts for all cost draws below the threshold  $\xi^* = S/w$  and does not adjust otherwise.*

**Definition 2** (Probability choice formulation).

*Given the payoff  $S$ , the following defines the probability choice payoff  $\Pi^{PC}(S)$  and the optimal adjustment probability  $\varphi^*$ : The agent chooses  $\varphi \in [0, 1]$ , paying a convex cost  $C(\varphi)$ :*

$$\Pi^{PC}(S) = \max_{\varphi \in [0, 1]} \{\varphi \cdot S - C(\varphi)\}, \quad (2)$$

*where  $C : [0, 1] \rightarrow \mathbb{R}_+$  with  $C(0) = 0$ ,  $C$  increasing and convex. The agent trades off the expected benefit of adjusting ( $\varphi \cdot S$ ) against the cost of choosing a high adjustment probability ( $C(\varphi)$ ).*

The duality theorem establishes that these two formulations are ex-ante equivalent: same value functions and same adjustment probabilities, with  $C$  derived from  $G$  via the quantile integral. For each distribution  $G$  with a positive density, there exists a unique convex cost function  $C$  with  $C(0) = 0$  that delivers this equivalence.

## 2.2 Main result

**Theorem 1** (Analytical convex duality of fixed costs).

*The fixed cost formulation (Definition 1) and the probability choice formulation (Definition 2) are equivalent under the convex cost function constructed in closed form by the quantile integral*

$$C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi). \quad (3)$$

*Specifically:*

- (i) **Payoff equivalence.**  $\Pi^{FC}(S) = \Pi^{PC}(S)$  for all  $S \geq 0$ . The two formulations are Fenchel conjugate duals:  $\Pi^{PC}(S)$  is the conjugate of  $C$ , and  $C(\varphi)$  is the conjugate of  $\Pi^{FC}$ .
- (ii) **Policy equivalence.** The optimal adjustment probability is the same:  $\varphi^* = G(S/w)$ . Moreover, if the probability choice model randomises by drawing  $\xi \sim G$  and adjusting whenever  $\xi \leq G^{-1}(\varphi^*)$ , the two models produce identical binary outcomes path by path.

*The construction (3) is the static, environment-free analytical link between the primitive distribution  $G$  and the dual convex cost  $C$ : it depends only on  $(G, w)$ , with the surplus  $S$  left abstract.*

*Proof.*

**(i) Payoff equivalence.** Substituting the threshold  $\xi^* = S/w$  (assuming  $S/w \leq \bar{\xi}$ ; for  $S/w > \bar{\xi}$ , both formulations yield  $\Pi(S) = S - wE[\xi]$  at  $\varphi^* = 1$ , and for unbounded support  $\varphi^* \in (0, 1)$  for every finite  $S$ ) into (1):

$$\Pi^{FC}(S) = S \cdot G(S/w) - w \int_0^{S/w} \xi dG(\xi). \quad (4)$$

The first term is the expected surplus from adjusting: the probability of adjustment  $G(S/w)$  times the surplus  $S$ . The second term is the expected cost paid by adjusters. Setting  $\varphi = G(S/w)$  so that  $S/w = G^{-1}(\varphi)$ :

$$\Pi^{FC}(S) = \varphi \cdot S - w \int_0^{G^{-1}(\varphi)} \xi dG(\xi) = \varphi S - C(\varphi).$$

This is the objective of (2) evaluated at  $\varphi^* = G(S/w)$ . Since  $C'(\varphi) = wG^{-1}(\varphi)$  is non-decreasing,  $C$  is convex, so the first-order condition  $S = C'(\varphi)$  is sufficient and  $\varphi^* = G(S/w)$  attains the supremum. The conjugate relationship follows from the definition of the Fenchel conjugate:  $\sup_{\varphi} \{\varphi S - C(\varphi)\} = \Pi^{PC}(S) = \Pi^{FC}(S)$ .



(ii) **Policy equivalence.** The first-order condition of (2) is  $S = C'(\varphi)$ . Differentiating (3) via the fundamental theorem of calculus gives  $C'(\varphi) = wG^{-1}(\varphi)$ , so  $G^{-1}(\varphi^*) = S/w$ , i.e.,  $\varphi^* = G(S/w)$  — the same threshold probability as in Definition 1. For the path-by-path equivalence: in the fixed cost model, the agent draws  $\xi \sim G$  and adjusts whenever  $\xi \leq S/w$ , which occurs with probability  $\varphi^*$ . If the probability choice model uses the same draw and adjusts whenever  $\xi \leq G^{-1}(\varphi^*)$ , the two rules coincide and the realised adjustment indicators are identical. The path-by-path coupling presupposes i.i.d. cost draws across time and agents; persistent-type heterogeneity in  $\xi$  preserves distributional but not trajectory equivalence. ■

In the fixed cost model, agents with low cost draws adjust and those with high draws do not. The expected payoff equals the adjustment probability times the surplus, minus the average cost paid by adjusters. In the probability choice model, the agent chooses an adjustment probability and pays a cost  $C(\varphi)$ . Proposition 1 below confirms that  $C$  is convex and characterises its shape. The theorem says these two descriptions yield exactly the same payoff and adjustment probability, provided  $C$  is constructed from  $G$  via (3).

The Fenchel conjugate structure places the two formulations on equal mathematical footing. The payoff function  $\Pi^{PC}(S)$  and the cost function  $C(\varphi)$  are conjugates of each other: the Fenchel conjugate of  $C$  yields  $\Pi^{PC}$ , and the conjugate of  $\Pi^{PC}$  yields  $C$ . The two express the same ex-ante value function in dual variables — one in the surplus  $S$ , the other in the adjustment probability  $\varphi$ . Any result proved for one variable translates immediately to the other. At the ex-ante value/probability level, the two are not different theories but different parametrisations of a single theory; under different primitive interpretations of  $G$  (physical cost versus information-processing penalty), the welfare and identification implications can differ, as discussed in Section 2.6. When  $G$  is uniform,  $C$  is quadratic and the equivalence can be verified by direct calculation (see the investment application below).

**Relation to discrete choice theory** As noted in the introduction, the Williams-Daly-Zachary (WDZ) theorem (McFadden, 1978, 1981) establishes that the expected maximum utility under random taste shocks is the Fenchel conjugate of a generating function determined by the shock distribution. For binary choice, the WDZ identity and Theorem 1 share the same algebraic skeleton (Table F.1 in Online Appendix F). The binary conjugate identity is therefore not new; the contributions here lie in what it enables — the distribution-theoretic taxonomy, the structural foundation for Khan-Thomas, and the continuous pricing spectrum — all of which are absent from the discrete choice literature. The duality is the tool; the taxonomy and the pricing application are the results.

Two further distinctions from WDZ are worth noting. First, the source of randomness



differs: in WDZ, randomness enters payoffs via taste shocks  $\varepsilon_j$ ; here, randomness enters costs via the draw  $\xi \sim G$ . For binary choice the two are isomorphic under a sign change. Second, the coupling is new. WDZ establishes equivalence of expected payoffs and choice probabilities; Theorem 1 further constructs a natural randomisation device under which the realised adjustment decisions coincide path by path — not merely in distribution.

**Relation to Alvarez-Lippi-Oskolkov (2022)** Alvarez, Lippi, and Oskolkov (2022) establish, in a continuous-time sticky-price setting with Brownian gap and Poisson cost arrivals, that any non-decreasing symmetric hazard  $\Lambda(x)$  admits two foundations: a random menu cost distribution  $G$  (their Section 2.1) and a convex flow cost  $c(\ell)$  on Poisson intensity (their Section 2.2), with  $c$  recovered from  $\Lambda$  by solving a second-order linear ordinary differential equation in the value function (their Lemma 1, Proposition 1). The convex-cost-over-adjustment-rate representation is therefore already present in their framework, and a direct calculation shows that on the bounded probability domain  $\varphi \in [0, 1]$  Theorem 1’s  $C(\varphi)$  coincides exactly with  $w$  times the ALO Section 2.2 cost  $c(\varphi)$ : both equal  $w \int_0^\varphi G^{-1}(s) ds$ .

Theorem 1 therefore does not claim the existence of the convex-cost representation as new. What it adds is three things. First, the construction is static and environment-free:  $C$  is the Fenchel conjugate of the gain function in a binary problem with random fixed cost  $\xi \sim G$ , depending only on  $(G, w)$  with the surplus  $S$  left abstract. ALO’s  $c$  is recovered within a specific dynamic pricing environment with parameters  $(r, B, \sigma^2)$ ; the explicit closed-form quantile-integral expression  $C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi)$  falls out as a corollary of working in this static frame. Second, the static frame makes the same object portable across applications: investment surplus, RI surplus, labour-supply surplus, etc. all admit the same construction without re-deriving the dual cost in each dynamic environment. Third, Theorem 1(ii) constructs a natural coupling under which the realised binary outcomes coincide path by path, not merely in distribution — a statement absent from ALO. The pricing application of Section 3 is thus one of several instantiations spanning menu costs, lumpy investment, rational inattention, discrete choice, and control costs, with the explicit closed-form construction inherited from the static formulation rather than re-derived in each setting. In the pricing instantiation, the power-law  $G$  of Section 3 induces a hazard  $\Lambda(x) \propto |x|^{2\gamma}$  — the power-hazard family ALO study, with exponent  $\nu = 2\gamma$ ; the cost-concentration parameter  $\gamma$  thus gives a structural reading of their hazard exponent.

**Relation to control costs** In the game theory literature, Mattsson and Weibull (2002) and Hofbauer and Sandholm (2002) study agents who choose mixed strategies subject to convex “control costs” that penalise departures from a default action. Deviating from habit-

ual behaviour is costly, and the cost increases as the deviation becomes more extreme. The probability choice formulation (2) has exactly this structure:  $C(\varphi)$  is the cost of choosing adjustment probability  $\varphi$  relative to the default of not adjusting ( $\varphi = 0$ ). Theorem 1 therefore shows that the fixed cost model *is* a control cost model. The control cost is not an ad hoc penalty imposed by the modeller. It is the endogenous cost function determined by the distribution of frictions that the agent faces.

## 2.3 Properties of the induced cost function

The induced cost function  $C(\varphi)$  inherits its shape entirely from the distribution  $G$ . The following proposition characterises its key properties and provides the economic interpretation that underlies all subsequent applications.

**Proposition 1** (Cost function properties).

*The induced cost function (3) satisfies:*

- (i)  $C(0) = 0$ ,  $C(1) = w E_G[\xi]$ .
- (ii)  $C'(\varphi) = w G^{-1}(\varphi)$ : *marginal cost equals the cost of the marginal adjuster.*
- (iii)  $C''(\varphi) = w/g(G^{-1}(\varphi)) > 0$ : *strictly convex whenever  $g > 0$ .*
- (iv)  $C$  is  $C^2$  if  $g$  is  $C^1$  and positive.

*Proof.* See Online Appendix A. ■

Part (ii) carries the core economic content. Agents are ranked by their cost draws: those with the lowest  $\xi$  adjust first. The marginal cost  $C'(\varphi) = w G^{-1}(\varphi)$  is the cost of the agent just indifferent at the current threshold — the  $\varphi$ -th quantile of the cost distribution. Each increment in  $\varphi$  requires inducing a more expensive agent to adjust, which is the force that makes  $C$  convex. This is the economic origin of convexity: it is not assumed but arises endogenously from the ranking of heterogeneous agents by their costs.

Part (iii) reveals how the density  $g$  governs the strength of this convexity. Where  $g$  is large, many agents are bunched near the threshold. A small increase in the surplus tips many of them into adjustment, so the marginal cost rises gently. Where  $g$  is small, few agents are near the threshold. The same increase tips few additional agents, and the marginal cost rises sharply. In short: crowded margins mean gentle curvature; sparse margins mean sharp curvature.

**Corollary 1** (Shape of  $G$  determines curvature).

Since  $C''(\varphi) = w/g(G^{-1}(\varphi))$  by Proposition 1(iii), the monotonicity of the density  $g$  across the support of  $G$  determines the monotonicity of  $C''$  as  $\varphi$  traverses  $(0, 1)$ . The three canonical cases are:

- (i)  $g$  increasing  $\Rightarrow C''$  decreasing: there are relatively more agents at higher cost quantiles, so inducing additional adjustment becomes progressively easier. Adjustment is “cheap at the margin” for high  $\varphi$ .
- (ii)  $g$  decreasing  $\Rightarrow C''$  increasing: agents thin out at higher cost quantiles, so inducing additional adjustment becomes progressively more expensive. Adjustment becomes disproportionately costly at high  $\varphi$ .
- (iii)  $g$  constant (uniform)  $\Rightarrow C''$  constant: agents are equally spaced throughout the cost distribution, yielding the standard quadratic cost function with constant curvature.

*Proof.* See Online Appendix A. ■

The following corollary applies the mapping (3) to four canonical distributions, recovering well-known cost function specifications as special cases of the single formula  $C(\varphi) = w \int_0^\varphi G^{-1}(u) du$ . Each distribution represents a different assumption about the cross-sectional heterogeneity of adjustment costs, and the induced cost function inherits its functional form from the shape of  $G$ .

**Corollary 2** (Canonical distributions).

Applying (3) to four canonical distributions recovers standard cost-function specifications in closed form:

- (i) **Uniform**  $G = U[0, \bar{\xi}]$ :  $C(\varphi) = \frac{w\bar{\xi}}{2}\varphi^2$ . The constant density produces constant curvature, so each unit increase in the adjustment probability costs the same marginal increment. This is the standard quadratic adjustment cost used throughout the macroeconomics and industrial organisation literatures.
- (ii) **Power law**  $G(\xi) = (\xi/\bar{\xi})^\gamma$ ,  $\gamma > 0$ :  $C(\varphi) = \frac{\gamma}{\gamma+1}w\bar{\xi}\varphi^{1+1/\gamma}$ . The cost function is iso-elastic with elasticity  $1+1/\gamma$ , and the shape parameter  $\gamma$  governs the curvature profile. For  $\gamma > 1$ , the density  $g(\xi) = \gamma\xi^{\gamma-1}/\bar{\xi}^\gamma$  is increasing: most cost draws are high, so cheap adjusters are scarce and  $C$  is less curved than quadratic. For  $\gamma < 1$ , the density is decreasing: most cost draws are low, cheap adjusters are abundant, and  $C$  is more curved than quadratic. The uniform case is  $\gamma = 1$ . As  $\gamma \rightarrow \infty$ , the density concentrates near  $\bar{\xi}$  and  $C$  approaches linearity (the degenerate limit). As  $\gamma \rightarrow 0$ , the density

concentrates near zero and  $C$  approaches zero uniformly on  $[0, 1]$  (free adjustment in the limit). The power law family thus spans the full range from lumpiness to frictionless adjustment, indexed by a single parameter. This family plays the central role in the pricing application of Section 3.

(iii) **Exponential**  $G(\xi) = 1 - e^{-\lambda\xi}$  on  $[0, \infty)$ :  $C'(\varphi) = -\frac{w}{\lambda} \ln(1 - \varphi)$ , giving  $C(\varphi) = \frac{w}{\lambda}[(1 - \varphi) \ln(1 - \varphi) + \varphi]$ . The logarithmic marginal cost diverges as  $\varphi \rightarrow 1$ , severely penalising near-certain adjustment. This reflects the exponential distribution's unbounded support: to guarantee adjustment with probability close to one, the agent must be willing to pay arbitrarily high cost draws, which makes the marginal cost of the last increment unbounded, even though the total expected cost  $C(1) = w/\lambda$  remains finite. As shown in Section 2.5, this cost function is the generalised KL divergence (I-divergence) from the no-adjustment prior, the Bregman divergence generated by  $h(x) = x \ln x$ .<sup>1</sup> It coincides with the mutual information cost in the binary rational inattention model under a degenerate default (Sims, 2003; Matějka and McKay, 2015).

(iv) **Degenerate** at  $\bar{\xi}$ :  $C(\varphi) = w\bar{\xi}\varphi$ . The cost function is linear, with no strict convexity. The only optimal choices are  $\varphi = 0$  (never adjust) or  $\varphi = 1$  (always adjust), depending on whether  $S$  falls below or above  $w\bar{\xi}$ . This is the canonical binary threshold rule of the deterministic fixed cost model — the singular case analysed in Proposition 2 below.

*Proof.* See Online Appendix A. ■

These are not four different models — they are four special cases of one model, distinguished solely by the shape of  $G$ .

## 2.4 Universality of the convex representation

The degenerate case — a single, known fixed cost — is the only setting in which the problem is not strictly convex. This section shows that this case is a knife-edge singularity: arbitrarily close perturbations of a point mass by absolutely continuous distributions with positive density restore strict convexity. The convex representation is the non-knife-edge case; the deterministic fixed cost is the singular exception.

**Proposition 2** (Universality).

Let  $\bar{\xi} > 0$  be a deterministic fixed cost. For any  $\epsilon \in (0, \bar{\xi})$ , perturb the cost to  $\xi \sim G_\epsilon \equiv$

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<sup>1</sup>Strict Shannon KL from a degenerate prior is infinite for any positive  $\varphi$ ; the cost here is the generalised KL / I-divergence, finite on the simplex interior.

$U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$ . The induced cost function is

$$C_\epsilon(\varphi) = w(\bar{\xi} - \epsilon)\varphi + w\epsilon\varphi^2, \quad (5)$$

which is strictly convex with  $C''_\epsilon(\varphi) = 2w\epsilon > 0$  for every  $\epsilon > 0$ . As  $\epsilon \rightarrow 0$ :

- (i)  $C_\epsilon(\varphi) \rightarrow w\bar{\xi}\varphi$  pointwise: the linear (degenerate) cost.
- (ii)  $C''_\epsilon(\varphi) \rightarrow 0$ : curvature vanishes.
- (iii) The optimal  $\varphi^*_\epsilon(S) = \max\left\{0, \min\left\{\frac{S-w(\bar{\xi}-\epsilon)}{2w\epsilon}, 1\right\}\right\}$  converges to  $\mathbf{1}[S \geq w\bar{\xi}]$ : the binary threshold rule.

Thus the deterministic fixed cost is the singular limit of a family of strictly convex problems.

*Proof.* For  $G_\epsilon = U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$ , applying (3) gives  $C_\epsilon(\varphi) = w(\bar{\xi} - \epsilon)\varphi + w\epsilon\varphi^2$ , with  $C''_\epsilon = 2w\epsilon > 0$ . The full derivation is in Online Appendix A.  $\blacksquare$

The result recasts the standard view that convex costs are a “smooth approximation” to the “true” lumpy friction. The deterministic fixed cost is the knife-edge limit; arbitrarily close perturbations restore strict convexity. The curvature  $C''_\epsilon = 2w\epsilon$  measures the “softness” of the adjustment margin: even an infinitesimally small  $\epsilon$  produces a strictly convex problem.

**Epistemic interpretation and continuity in  $G$**  The deterministic fixed cost requires the agent to know the adjustment cost with infinite precision — an idealisation rarely satisfied in practice (staff time, managerial attention, installation logistics). Any  $\epsilon$ -band of noise makes the decision problem smooth and convex. The deterministic case is fragile under perturbation; the adjustment-probability representation varies continuously under standard regularity on  $G$  ( $g$  continuous and bounded away from zero on the relevant range). The set of distributions generating strictly convex  $C$  is dense under the weak topology: every neighbourhood of a point mass contains absolutely continuous distributions with strictly convex induced costs (Online Appendix A).

**Mean-preserving spreads and convexity** Proposition 2 shows that any perturbation of a point mass restores strict convexity. The next result records the integrated quantile order implied by MPS.

**Proposition 3** (Dispersion and convexity).

*Let  $G_1$  and  $G_2$  be distributions on  $[0, \bar{\xi}]$  with the same mean and let  $G_2$  be a mean-preserving spread of  $G_1$ . The induced costs satisfy:*

- (i)  $C_1(1) = C_2(1) = w E[\xi]$ .
- (ii)  $C_2(\varphi) \leq C_1(\varphi)$  for all  $\varphi \in [0, 1]$ : greater dispersion lowers the cost function in the interior, via the integrated quantile order  $\int_0^t G_2^{-1} \leq \int_0^t G_1^{-1}$  characterising MPS.
- (iii) Within the power law family, the marginal cost schedules  $C'_\gamma(\varphi) = w\bar{\xi}\varphi^{1/\gamma}$  rotate around a single  $\varphi_0 \in (0, 1)$  as  $\gamma$  varies. Outside parametric families, multiple crossings of  $C'_2 - C'_1$  are possible without an additional single-crossing assumption.

The degenerate distribution maximises  $C$  on  $(0, 1)$  and is the unique case in which  $C$  is linear; any non-degenerate  $G$  with positive density makes  $C$  strictly convex.

*Proof.* See Online Appendix A. ■

## 2.5 Fixed costs as generalised entropies

This subsection develops the  $C$ -side correspondence: for researchers whose primitive is a convex penalty  $C$  on choice probabilities, the duality reveals which underlying distribution  $G$  they have implicitly assumed. The cost function  $C(\varphi)$  is fully determined by  $G$  and inherits its shape from the quantile function  $G^{-1}$ . Canonical distributions  $G$  yield cost functions that coincide with well-known information-theoretic entropies. The implication is that the choice of cost specification in any model with frictional adjustment is, implicitly, an assumption on the underlying distribution  $G$  of fixed costs.

**Proposition 4** (Entropy correspondence).

*The quantile integral (3) converts each canonical distribution  $G$  into a cost function with a known information-theoretic counterpart. Three correspondences follow:*

- (i) **Exponential distribution:**  $G(\xi) = 1 - e^{-\lambda\xi}$  on  $[0, \infty)$ . Then

$$C'(\varphi) = -\frac{w}{\lambda} \ln(1 - \varphi), \quad C(\varphi) = \frac{w}{\lambda} [(1 - \varphi) \ln(1 - \varphi) + \varphi].$$

*The adjustment probability under exponential costs is  $\varphi^* = 1 - e^{-\lambda S/w}$ , the CDF of the exponential distribution evaluated at  $S/w$ . Note that this is not the logistic function  $1/(1 + e^{-S/\sigma})$  that arises in logit discrete choice; the two share a sigmoid shape but differ in functional form. The true logit correspondence arises in the multi-action extension (Section 2.8), where i.i.d. Gumbel-distributed costs across alternatives yield the multinomial logit.*

(ii) **Power law distribution:**  $G(\xi) = (\xi/\bar{\xi})^\gamma$  on  $[0, \bar{\xi}]$ . Then

$$C(\varphi) = \frac{\gamma}{\gamma + 1} w \bar{\xi} \varphi^{1+1/\gamma}.$$

This is proportional to  $\varphi^q$  with  $q = 1 + 1/\gamma$ , which shares the power-law exponent with the Tsallis  $q$ -divergence from the no-adjustment prior (Tsallis, 1988). The parameter  $\gamma$  controls  $q$ :  $\gamma = 1$  gives  $q = 2$  (quadratic). As  $\gamma \rightarrow \infty$ ,  $q \rightarrow 1$  and  $C(\varphi) \rightarrow w \bar{\xi} \varphi$ , a linear function. Recovering the logarithmic (Shannon) form requires the exponential distribution, not the power law limit.

(iii) **Uniform distribution:**  $G = U[0, \bar{\xi}]$ . Then  $C(\varphi) = \frac{w \bar{\xi}}{2} \varphi^2$ , a quadratic penalty. This is the simplest and most widely used convex cost specification, and it coincides with the  $L^2$  penalty that appears in many regularisation frameworks.

*Proof.* See Online Appendix A. ■

The correspondences are of functional form, not of economic mechanism: exponential  $G$  yields the KL divergence used in binary rational inattention (Sims, 2003; Matějka and McKay, 2015); uniform  $G$  yields the quadratic attention penalty of Gabaix (2014); in the multi-action extension (Section 2.8), i.i.d. Gumbel costs yield negative Shannon entropy and the multinomial logit. The reduced-form  $C(\varphi)$  is the same; welfare and identification implications, taken up in Section 2.6, differ.

**The distribution  $G$  as a Rosetta Stone** Table 1 summarises the mapping. Five literatures — menu costs, lumpy investment, discrete choice, rational inattention, and control costs — have independently developed cost specifications that are the parametric image of an implicit assumption on  $G$ . Making this assumption explicit converts a free modelling choice into a parametric restriction on a cost distribution that can be measured.

## 2.6 Welfare content of the duality

The welfare cost of adjustment at a given equilibrium is  $C(\varphi^*)$ . The identity has two uses: a smooth closed-form welfare measure for fixed cost models, and a physical-cost interpretation for the abstract penalty functions of rational inattention and discrete choice.

**From fixed to convex: smooth welfare analysis** The probability choice formulation replaces  $\max\{S - w\xi, 0\}$  with the concave objective  $\max_\varphi\{\varphi S - C(\varphi)\}$ , with smooth first-order condition  $S = C'(\varphi) = wG^{-1}(\varphi)$  and policy function  $\varphi^* = G(S/w)$ . This delivers a



Table 1: The distribution  $G$  as a Rosetta Stone

| Distribution $G$        | Cost $C(\varphi)$                      | Form         | Entropy           | Literature                  |
|-------------------------|----------------------------------------|--------------|-------------------|-----------------------------|
| Exponential             | $(1-\varphi) \ln(1-\varphi) + \varphi$ | Log. MC      | KL divergence     | Rational inattention        |
| Power law ( $\gamma$ )  | $\varphi^{1+1/\gamma}$                 | Iso-elastic  | Tsallis $q$ -div. | Generalised discrete choice |
| Uniform ( $\gamma=1$ )  | $\varphi^2$                            | Quadratic    | $L^2$             | Sparsity                    |
| Degenerate              | $\xi \varphi$ on $\{0, 1\}$            | Linear       | —                 | Lumpy adjustment            |
| Type I EV (multinomial) | $\sum \varphi_j \ln \varphi_j$         | Neg. entropy | Shannon (mv)      | Logit discrete choice       |

*Notes:* Each row specifies a distribution  $G$  of fixed costs. The second column gives the induced convex cost function  $C(\varphi)$  via equation (3) (up to scale). The fourth column identifies the information-theoretic entropy to which  $C$  corresponds. The last column indicates the literature where this cost function appears. “Log. MC” = logarithmic marginal cost. “mv” = multivariate. The first four rows are binary choice; the last row is the multi-action extension (Proposition 5).

closed-form welfare measure for fixed-cost models. For the power law family, all aggregate objects follow in closed form:  $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ ,  $\kappa = (1 + 2\gamma)\bar{\Lambda}$  (exact when  $f_{ss}$  is supported in the interior of the inaction band; approximate otherwise — Corollary 6), and the welfare cost  $C(\varphi^*)$ . The total welfare cost at a given equilibrium is

$$\text{Welfare cost} = w \int_0^{G^{-1}(\varphi^*)} \xi dG(\xi) = C(\varphi^*), \quad (6)$$

a compact representation given  $\varphi^*$ . Note that  $\varphi^*$  itself still depends on  $G$  through the full equilibrium, so (6) summarises welfare without shortcutting the equilibrium computation.

**From convex to fixed: welfare interpretation of information costs** In rational inattention, the mutual-information cost  $\lambda \sum \varphi_j \ln \varphi_j$  is measured in nats;  $\lambda$  is the shadow price of attention with no independent resource interpretation. Under the fixed-cost primitive in the binary exponential- $G$  case,  $C(\varphi^*)$  equals the expected physical cost paid by adjusters — and  $\lambda$  becomes the rate parameter of the cost distribution, in principle estimable from micro data. The resource-cost reading applies only under non-negative physical-cost primitives; the Gumbel/multi-action logit case (Proposition 5(ii)) is a reduced-form conjugacy with  $\xi$  a real-valued payoff shock and does not carry the resource interpretation.

## 2.7 Theoretical and computational implications

This subsection develops the  $G$ -side payoff: for researchers whose primitive is a cost distribution  $G$ , the duality delivers a smooth dynamic value function, a closed-form first-order condition, and direct integration into existing FOC-based solvers. In dynamic problems, the duality applies state by state. Consider a Bellman with extensive choice  $p \in [0, 1]$  (adjust or

not) and intensive choice  $x'$  (the post-adjustment target).

**Theorem 2** (Differentiability of the lumpy adjustment value function).

*Consider the dynamic problem of Section 3.1, in adjustment / non-adjustment branch form with values  $v^a, v^n$ , per-period payoffs  $F^a, F^n$ , and convex cost  $C$  (full statement of the generic branch Bellman in Online Appendix G). Assume:*

- (A1)  *$G$  is absolutely continuous on  $[0, \bar{\xi}]$  with density  $g \in C^1$  and  $g > 0$  on the interior of its support.*
- (A2) *The per-period payoffs  $F^a(\cdot, \cdot; X)$  and  $F^n(\cdot; X)$  are bounded and  $C^1$ , with  $F^a$  moreover  $C^2$  in the intensive target  $x'$  and strictly concave in  $x'$ ; the transition kernel and the stochastic discount factor  $m(X, X')$  are  $C^1$  in their continuous arguments.*
- (A3) *The state spaces for  $x$  and  $x'$  are compact. Along the iteration in the proof, the full objective of the resetting firm,*

$$F^a(x', x; X) + \beta \mathbb{E} m(X, X') v(x'; X'),$$

*is strictly concave in  $x'$  with a unique interior optimum  $x'^*(x; X)$  and a non-degenerate second-order condition, uniformly in  $(x, X)$ . In primitive terms:  $F^a$  is  $C^2$  in  $x'$  with curvature bounded away from zero,  $F^a_{x'x'} \leq -\underline{\sigma} < 0$  for all  $(x', x, X)$ ; the cross-partial  $F^a_{x'x}, F^a_{xx}$  are uniformly bounded; and  $\beta$  is small enough that the continuation does not overturn this curvature.<sup>2</sup>*

*Then the Bellman operator has a unique fixed point  $v(\cdot; X)$  in the space of bounded measurable functions, which is moreover  $C^1$  in  $x$  for each  $X$ , with  $v_x$  given by the Benveniste-Scheinkman envelope.*

*Proof sketch.* By (A1) and Proposition 1, the Fenchel conjugate

$$c^*(S; X) \equiv \max_{p \in [0,1]} \{pS - C(p; X)\}$$

is  $C^1$  in  $S$ . Concentrating out the extensive choice reduces the Bellman to  $v(x; X) = v^n(x; X) + c^*(S(x; X); X)$ , with  $S = v^a - v^n$  and  $v^a$  containing the inner max over  $x'$ . Existence of a unique fixed point in bounded measurable functions follows from Blackwell sufficient conditions (Stokey, Lucas, and Prescott 1989, Ch. 3). For  $C^1$  regularity, I verify

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<sup>2</sup>(A3) is the joint compactness, interior-optimum, and strict-second-order-condition hypothesis of Santos (1991), Theorem 4.1, stated on the full objective rather than on the per-period payoff alone. Strict concavity of  $F^a$  by itself does not suffice: in principle the continuation term could undo it. (A3) rules this out.

that the Bellman operator preserves the class of bounded  $C^1$  functions. Take a  $C^1$  candidate continuation  $w$ . The adjustment-branch maximisation  $\max_{x'} \{F^a(x', x; X) + \beta \mathbb{E} m w(x'; X')\}$  satisfies the hypotheses of Santos (1991), Theorem 4.1, under (A2)–(A3): compact state-action space, interior intensive optimum, and uniformly strict concavity of the *full maximand* (not just  $F^a$ ) along the iteration. Its value is therefore  $C^1$  in  $x$ , with derivative given by the Benveniste-Scheinkman envelope (Benveniste and Scheinkman, 1979). The non-adjustment branch is  $C^1$  by direct differentiation; composition with the smooth  $c^*$  then yields a  $C^1$  iterate, and the fixed point inherits  $C^1$  in  $x$  for each  $X$ . The substantive content of (A3) is that the continuation curvature, weighted by  $\beta$ , cannot overturn  $F^a$ ’s primitive concavity along the iteration; this is the standard joint hypothesis under which Santos’ theorem applies. The full argument is in Online Appendix G.

The contribution of the duality at this step is to make the smoothness of the ex-ante value operational. The  $C^1$  regularity itself is standard, following from general dynamic-programming arguments (Santos 1991) and obtained in lumpy settings via smooth-pasting at inaction-region boundaries (Balei and Blanco 2021). The cost object coincides with that of Alvarez, Lippi, and Oskolkov (2022), recovered there from the hazard by solving a second-order linear ordinary differential equation in the value function. *What is new here is the route:* the map from the cost distribution to the convex cost,  $G \mapsto C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi)$ , is environment-free, depending only on  $(G, w)$  with the surplus  $S$  left abstract, and it yields a closed-form first-order condition  $S = C'(\varphi)$  in any environment admitting a scalar adjustment surplus. Methods that work with the realised threshold rule (or the Ss-band cutoff in  $x$  for given  $\xi$ ) face a kink in the conditional value function and require threshold-specific machinery; smooth-pasting approaches handle this at the inaction-region boundaries through optimal-control conditions; ODE-based approaches construct  $v$  from a Sturm-Liouville-type solution. The Fenchel-dual reformulation supplies a smooth maximand globally — the convex cost  $C$  derived from the absolutely continuous  $G$  via Proposition 1 — to which Santos’ theorem applies directly. This bridge, not the  $C^1$  conclusion, is what Theorem 2 contributes. ■

**Corollary 3** (Policy differentiability).

*Under (A1)–(A3), the intensive policy  $x^*(x; X)$  is  $C^1$  in  $x$  for each  $X$ , and the extensive policy  $p^*(x; X) \in [0, 1]$  is continuous on  $(x, X)$  and  $C^1$  on the interior set  $\{(x, X) : p^*(x; X) \in (0, 1)\}$ .*

*Proof.* The joint objective in  $(p, x')$  is *not* jointly concave: the coupling  $p \mathbb{E} m v(x'; X')$  entering the Bellman is bilinear in  $(p, x')$ , so strict convexity of  $C$  in  $p$  together with strict concavity of  $F^a$  in  $x'$  does not imply joint concavity. The two margins are therefore treated

separately. The intensive choice solves the inner maximisation defining  $v^a$ , which is independent of  $p$ ,  $C^1$  in  $x'$ , and strictly concave in  $x'$  by (A2); the first-order condition has a unique solution  $x'^*$ , and the implicit function theorem upgrades it to  $C^1$  using the strict SOC in (A3) and  $v \in C^1$  from Theorem 2. The extensive choice solves the static problem  $\max_{p \in [0,1]} \{p S(x; X) - C(p; X)\}$ , where  $S = v^a - v^n$  is the surplus;  $S$  is  $C^1$  in  $x$  (Theorem 2), and strict convexity of  $C$  in  $p$  (from (A1)) yields a unique smooth maximiser  $p^*$  via  $C_p(p^*; X) = S(x; X)$ , with  $p^* \in C^1$  on the interior of  $\{p^* \in (0, 1)\}$  by the implicit function theorem. The full argument is in Online Appendix G. ■

Standard equilibrium-existence and comparative-statics arguments apply directly.

A short clarification on what the duality does and does not change at this step. With continuous  $G$ , the ex-ante integrated payoff  $\Pi^{FC}(S) = \int \max\{S - w\xi, 0\} dG(\xi)$  already equals the Fenchel conjugate  $c^*(S)$  and is smooth in  $S$ . The duality's contribution is to expose this smoothness as a closed-form FOC  $S = C'(\varphi)$ , rather than as an implicit integration over  $\xi$ , and to replace the discontinuous realised adjustment indicator  $\mathbf{1}[\xi \leq S/w]$  by the smooth ex-ante probability  $\varphi^* = G(S/w)$ . Methods that track the realised threshold rule (or the Ss-band cutoff in  $x$  for a given  $\xi$ ) face a kink in the conditional value function and require threshold-specific machinery; the FOC representation eliminates that machinery. The ex-ante value itself loses smoothness only in the degenerate- $G$  (Goloso-Lucas) limit, where  $\Pi^{FC}$  acquires a corner at  $S = w\bar{\xi}$  and  $\varphi^* \in \{0, 1\}$ .

**What Theorem 2 is for** The  $C^1$  conclusion is not pursued here for its own sake. Its purpose is to make the dynamic model of Section 3.1 solvable globally, across the entire menu-cost–Calvo spectrum, without log-linearisation. Three uses, all exploited by the routine of Section 4.5. First, the regularity legitimises the Benveniste-Scheinkman envelope update  $V_x = (1 - \Lambda)V_x^n$  each period, so the resetter's intensive target  $x'^*$  is recovered from a smooth FOC against the continuation envelope rather than by an individual-level grid search. Second, it keeps matched-period continuation values continuous on the gap grid, which is what allows the scalar summary of the cross-section (the lagged price-dispersion index) to identify periods along the simulated equilibrium path. Third, it lets the routine work in the FOC–envelope representation continuously across the spectrum, with the Goloso-Lucas limit treated as the high- $\gamma$  endpoint at which the ex-ante value acquires a corner at the inaction-band boundary — a limit point the routine approaches continuously rather than a singularity that has to be handled separately. The comparison with smooth-pasting and ODE routes above is about how the  $C^1$  conclusion is reached; this paragraph is about what it is then used for.

**Computational routine** With  $v \in C^1$  (Theorem 2) and differentiable policies (Corollary 3), the routine proceeds in closed FOC–envelope form. The Fenchel step gives the smooth extensive-margin condition  $S = C'(p)$  in the interior, with a KKT projection at the boundary; the intensive choice satisfies its own FOC against the envelope of the continuation value. Standard perturbation methods apply without special treatment of the threshold, slotting the model into FOC-based heterogeneous-agent toolkits (Reiter, 2009; Auclert, Rognlie, and Straub, 2020); the intensive target  $x^{I*}$  is recovered from the FOC–envelope system without an individual-level grid search, in the endogenous-grid-like style of Carroll (2006). The same regularity also supports global nonlinear solution. The smoothness of  $v$  in  $x$  from Theorem 2 keeps matched-period continuation values continuous on the gap grid. In the NK calibration of Section 4, the price-dispersion index passes the within-exogenous-state monotonicity test of Lee (2026) as a scalar summary of the aggregate state. The repeated transition method then produces fully nonlinear impulse responses without parameterising laws of motion or linearising around steady state. The full routine is in Online Appendix G.

## 2.8 Beyond binary choice

The duality extends to multi-action settings via the same conjugate construction. Suppose the agent chooses among  $\{a_0, a_1, \dots, a_J\}$  ( $a_0$  = inaction), with  $a_j$  ( $j \geq 1$ ) yielding surplus  $S_j$  at random cost  $w\xi_j$ ,  $\xi_j \sim G_j$  independent across actions, and selects the action with the highest net payoff.

**Proposition 5** (Multi-action duality).

Let  $J \geq 1$  and suppose  $\xi_j \sim G_j$  independently for  $j = 1, \dots, J$ , with each  $G_j$  absolutely continuous with positive density on  $[0, \bar{\xi}_j]$ . Define  $S_0 = 0$ ,  $\xi_0 = 0$ . Then:

(i) The expected payoff  $\Pi^{FC}(\mathbf{S}) = E[\max_{j \in \{0, \dots, J\}} \{S_j - w\xi_j\}]$  equals

$$\Pi^{PC}(\mathbf{S}) = \max_{\boldsymbol{\varphi} \in \Delta^J} \sum_{j=0}^J \varphi_j S_j - C(\boldsymbol{\varphi}), \quad (7)$$

where  $C : \Delta^J \rightarrow \mathbb{R}$  is jointly convex and determined by the joint distribution of  $(\xi_1, \dots, \xi_J)$ , which under independence is pinned down by the marginals  $(G_1, \dots, G_J)$ .

(ii) (Payoff-shock analogue.) Reinterpret  $\xi$  as a real-valued payoff shock rather than a non-negative physical cost (so  $\eta_j \equiv -w\xi_j$  plays the role of a Gumbel taste shock in the standard random-utility setup). When  $\eta_j \stackrel{i.i.d.}{\sim} \text{Gumbel}(0, \sigma)$  for all  $j \geq 0$ , the conjugate construction yields  $C(\boldsymbol{\varphi}) = \sigma \sum_{j=0}^J \varphi_j \ln \varphi_j \leq 0$  (negative Shannon entropy on  $\Delta^J$ ).

The optimal choice probabilities are

$$\varphi_j^* = \frac{e^{S_j/\sigma}}{\sum_{k=0}^J e^{S_k/\sigma}}, \quad (8)$$

the multinomial logit of [McFadden \(1974\)](#).

*Proof.*

Apply the Fenchel conjugate construction to the multi-alternative maximum and specialise to the Gumbel distribution. The detailed proof is given in Online Appendix A. ■

When the surplus depends on a state  $x$ , the duality applies state by state:  $\varphi^*(x) = G(S(x)/w)$  is the endogenous state-dependent hazard. In dynamic settings the probability-choice formulation gives an explicit smooth first-order condition for the reseter; implementations that work with the realised threshold rule face a kink in the conditional value function at the adjustment threshold, and the degenerate- $G$  (Golosov-Lucas) limit produces a corner kink even in the ex-ante value.

## 2.9 Application to investment

The duality has a direct implication for the lumpy investment literature. In the [Thomas \(2002\)](#) model, [Gourio and Kashyap \(2007\)](#) show that the shape of the fixed cost distribution is the key determinant of whether lumpy micro-level investment spikes matter for aggregates. The duality makes this precise: each  $G$  maps to a unique convex cost  $C$ , and the shape of  $G$  governs the curvature of  $C$  and hence the aggregate investment elasticity. Under uniform  $G$ ,  $C$  is quadratic (Corollary 2), so each firm’s investment problem is individually convex — providing a partial-equilibrium foundation for the irrelevance result of [Khan and Thomas \(2008\)](#). The GE channel (endogenous factor price adjustment dampening investment spikes) provides another route to smoothness; the PE convexity established here provides an independent sufficient condition that does not rely on GE feedbacks. More broadly, any  $G$  with a positive density delivers smoothness at the individual level. The dispersion of  $G$  provides a continuous index of lumpiness: concentrated  $G$  (high  $\gamma$ ) means costs cluster near  $\bar{\xi}$  (similar across firms), so adjustment occurs in sharp spikes when surpluses cross the common threshold; dispersed  $G$  (low  $\gamma$ ) spreads costs widely, producing smoother adjustment. As dispersion shrinks to a point mass, the model recovers the binary threshold rule. [Oskolkov and Lippi \(2026\)](#) recover  $G$  from plant-level investment data, confirming that the shape of the cost distribution varies across sectors. Formal conditions for smoothness and the investment elasticity formula are in Online Appendix B.

### 3 The menu cost–Calvo spectrum

The deepest application of the duality is to price setting, where the distinction between time-dependent (Calvo) and state-dependent (menu cost) pricing is central to New Keynesian economics. The idea that these models sit on a spectrum is not new: [Dotsey, King, and Wolman \(1999\)](#) draw menu costs from a distribution, and [Caballero and Engel \(1993\)](#) parametrise the adjustment hazard directly. What the duality plus the power-law parametrisation add is a closed-form analytical decomposition: the repricing hazard  $\Lambda(x)$  is derived from  $G$  rather than assumed, and the frequency-selection decomposition is indexed by a single parameter. The distribution  $G$  governs the repricing hazard, the selection effect, the Phillips curve slope, and the degree of monetary non-neutrality.

This section follows an analytical strategy: replace the dynamic repricing surplus by its quadratic approximation  $S(x) = Bx^2$  and derive everything — the hazard, the Phillips curve slope  $\kappa = (1 + 2\gamma)\bar{\Lambda}$ , the decomposition into frequency and selection — in closed form. The choice is for clarity, not necessity. The full dynamic model (9)–(14) is itself solvable globally, and Section 4.5 does so. The reason to lead with the analytical approach is that closed-form expressions show which primitive moves which aggregate margin — a mapping that survives recalibration, and that a numerical solution at a single point of the parameter space cannot deliver on its own. The global solution then plays a complementary role: it checks numerically how well the closed-form decomposition approximates the unapproximated dynamic model. The dynamic surplus is only approximately quadratic, so  $\kappa = (1 + 2\gamma)\bar{\Lambda}$  holds to that approximation, not as an exact identity.

#### 3.1 The dynamic pricing problem

The application embeds the duality in a dynamic New Keynesian environment. I summarise the firm block here; the household block, the law of motion for the price-gap distribution, the price aggregator, and monetary policy are the standard New Keynesian closure and are stated in full in Online Appendix G. A monopolistically competitive firm has current log price  $p$  and desired log price  $p^*$ , with *price gap*  $x \equiv p^* - p$ . A non-adjusting firm’s gap drifts by  $-\pi'$  each period through aggregate inflation; an adjusting firm resets it to the optimal target.<sup>3</sup> Let  $X$  denote the aggregate state (the cross-sectional gap distribution and

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<sup>3</sup>Idiosyncratic shocks to the desired price can be added. They superimpose diffusion on the gap process and shape the cross-section, but leave the duality, the hazard  $\Lambda(x) = G(Bx^2/w)$ , and the frequency-selection decomposition unchanged. Crucially, the differentiability of the value function (Theorem 2) comes from the convexity of the Fenchel-dual cost  $C$ , not from idiosyncratic smoothing of the value function as in the menu-cost tradition. The cross-section is therefore kept non-degenerate by trend inflation alone, as in the global computation of Section 4.5 (with implementation details in Online Appendix G). Including idiosyncratic



the exogenous shocks),  $R(x; X)$  the per-period real profit at gap  $x$ ,  $\beta$  the discount factor, and  $m(X, X')$  the stochastic discount factor.

Each period the firm chooses a repricing probability  $\Lambda \in [0, 1]$ , paying the convex cost  $C(\Lambda)$  that the duality (Theorem 1) assigns to the menu-cost distribution  $G$ . Its value function solves

$$V(x; X) = \max_{\Lambda \in [0, 1]} \left\{ \Lambda V^a(X) + (1 - \Lambda) V^n(x; X) - C(\Lambda) \right\}, \quad (9)$$

$$V^a(X) = \max_{x'} \left\{ R(x'; X) + \beta \mathbb{E} \left[ m(X, X') V(x' - \pi'; X') \right] \right\}, \quad (10)$$

$$V^n(x; X) = R(x; X) + \beta \mathbb{E} \left[ m(X, X') V(x - \pi'; X') \right], \quad (11)$$

where  $V^a(X)$  is the value of repricing (resetting the gap to the optimal target  $x'^*$ ) and  $V^n(x; X)$  the value of not repricing. The *repricing surplus* is

$$S(x; X) \equiv V^a(X) - V^n(x; X), \quad (12)$$

and the duality's first-order condition  $S = C'(\Lambda)$  delivers the state-dependent hazard in closed form,

$$\Lambda(x; X) = G\left(\frac{S(x; X)}{w}\right) = \min \left\{ \left( \frac{S(x; X)}{w\xi} \right)^\gamma, 1 \right\}, \quad (13)$$

the second equality holding for the power law  $G(\xi) = (\xi/\bar{\xi})^\gamma$ . By Theorem 2 and Corollary 3,  $V$  is  $C^1$  in  $x$ , so the reset target solves the smooth first-order condition

$$\begin{aligned} R_x(x'^*; X) + \beta \mathbb{E} \left[ m(X, X') V_x(x'^* - \pi'; X') \right] &= 0, \\ V_x(x; X) &= (1 - \Lambda(x; X)) V_x^n(x; X), \end{aligned} \quad (14)$$

the second equation being the standard envelope condition (Benveniste and Scheinkman, 1979). There is no threshold to locate and no firm-level grid search: the entire policy reduces to a smooth first-order condition, which is what lets the model slot into standard heterogeneous-agent solvers. The dynamic Bellman (9)–(14) above is used in two places in what follows. The next subsection replaces the dynamic surplus by its quadratic approximation and derives the closed-form analytical results of this section from there. Section 4.5 returns to the full dynamic system and solves it globally, both as a check on the analytical approximation and to compute state-dependent impulse responses that a closed-form approach cannot.

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shocks does not change the results.

### 3.2 Static surplus specialisation

For the analytical results that follow I specialise the repricing surplus (12) to its second-order approximation around the optimum. To second order the gain from closing the gap is quadratic, and I adopt this static form for analytical clarity:

$$S(x) = Bx^2, \tag{15}$$

where  $B > 0$  depends on the demand elasticity and the discount factor.<sup>4</sup> The quadratic form reflects the approximate symmetry of profits around the optimum. The loss from mispricing grows with the square of the price gap: a firm with a 2% gap loses roughly four times as much as one with a 1% gap. The static Fenchel mapping is exact under this quadratic specialisation together with the interior-support condition stated below; Section 4.5 solves the dynamic model (9)–(14) globally and finds the decomposition to be a tight approximation to the unapproximated dynamic model (see below and Online Appendix G).

The firm draws a menu cost  $\xi \sim G$  each period and reprices if and only if the gain from repricing exceeds the cost  $w\xi$ , where  $w > 0$  is the wage (as in Section 2):

$$\text{Reprice if and only if } w\xi \leq Bx^2. \tag{16}$$

**What the dynamic surplus would change, and what it would not** In the full dynamic problem of Section 3.1, the repricing surplus is  $S(x; X) = V^a(X) - V^n(x; X)$  — the gain from resetting the price gap relative to leaving it at  $x$ . This dynamic surplus differs from the static  $Bx^2$  in two ways. First, it includes the option value of delaying: a firm that does not reprice today carries its mispricing forward, which the static surplus ignores. Second, its curvature accumulates the firm’s discounted future flow losses over its expected mispricing horizon, so the dynamic  $B$  is larger than the static  $\varepsilon/2$  by a factor that depends on the discount factor and the expected adjustment frequency. The dynamic surplus is, however, still approximately quadratic near the optimum — which is all the aggregate decomposition needs. The selection ratio  $\kappa/\bar{\Lambda} = 1 + 2\gamma$  (Corollary 6) does not depend on the level of  $B$ , so the same closed-form decomposition applies with the dynamic  $B$  in place of the static one (Alvarez, Le Bihan, and Lippi, 2016). The duality itself (Theorem 1) holds for any surplus function  $S$ ; specialising to a quadratic surplus is for clarity, not a restriction of the framework.

Section 4.5 checks this numerically. The dynamic model (9)–(14) is solved globally on a

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<sup>4</sup>This is standard; see Golosov and Lucas (2007) and Alvarez, Le Bihan, and Lippi (2016). The quadratic loss approximation is  $S(x) \approx \frac{\varepsilon}{2}x^2$  where  $\varepsilon$  is the demand elasticity, up to discounting.

simulated equilibrium path. The cross-section is generated by trend inflation alone, with no idiosyncratic price shocks. Three findings emerge.

(i) The dynamic surplus is approximately quadratic near the optimum, so the hazard keeps its power-law shape. The dynamic curvature  $B$  comes in roughly four times its static flow value, as the option-value argument above predicts.

(ii) The calibrated stationary distribution  $f_{ss}$  stays strictly inside the inaction band across the  $\gamma$ -spectrum, with negligible mass at the adjustment boundary. The boundary correction to  $\kappa$  is therefore numerically negligible.

(iii) Within each exogenous state, the resetter’s value and policy are monotone in the price-dispersion index  $\Delta$  — the within-exogenous-state monotonicity that the RTM matching method exploits.

Because  $\kappa/\bar{\Lambda}$  does not depend on  $B$ , the higher dynamic curvature leaves the decomposition — and the 44% selection share of Section 4 — numerically intact. The closed-form decomposition is thus a tight numerical approximation to the full dynamic model, with residual error governed by how far the dynamic surplus departs from exact quadraticity.

### 3.3 The repricing hazard

By Theorem 1, the firm’s repricing decision under the fixed cost formulation (Definition 1) is equivalent to the probability choice formulation (Definition 2) with convex cost  $C(\varphi)$ . The first-order condition  $Bx^2 = C'(\varphi) = wG^{-1}(\varphi)$  yields the optimal repricing probability as a function of the price gap:

**Definition 3** (Repricing hazard).

*The repricing hazard — the probability that a firm with price gap  $x$  adjusts its price — is*

$$\Lambda(x) \equiv G\left(\frac{Bx^2}{w}\right). \quad (17)$$

This is the central object of the pricing application. It is not a numerical output of a dynamic program; it is an explicit formula, given directly by the duality and the choice of  $G$ .<sup>5</sup> The repricing hazard  $\Lambda(x)$  summarises everything about the firm’s pricing behaviour that matters for aggregate dynamics. Conditional on  $(B, w)$  and the steady-state gap distribution  $f_{ss}$ , all aggregate pricing implications — inflation, the Phillips curve slope, the selection

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<sup>5</sup>The repricing hazard  $\Lambda(x)$  is the “adjustment function” in the language of Caballero and Engel (1993, 1999). Those papers take the adjustment function as a primitive and characterise how its shape determines aggregate dynamics. The present contribution is to derive  $\Lambda(x)$  from first principles as a function of the menu cost distribution  $G$ , establishing  $G$  as the deeper structural primitive.

effect, monetary non-neutrality — flow through  $\Lambda$ . The distribution  $G$  is therefore the structural primitive from which aggregate pricing dynamics are derived.

**Proposition 6** (Hazard properties).

*The repricing hazard  $\Lambda(x)$  satisfies:*

- (i)  $\Lambda(0) = G(0) = 0$ : *a firm at its optimal price never reprices. The gain from doing so is zero, and every cost draw is positive.*
- (ii)  $\Lambda$  *is symmetric in  $x$  and increasing in  $|x|$ : the repricing probability depends only on the magnitude of the gap, not its sign. The surplus  $Bx^2$  depends only on  $x^2$ . Larger gaps produce larger surpluses, which exceed more cost draws.*
- (iii)  $\Lambda'(x) = g(Bx^2/w) \cdot 2Bx/w$ : *the sensitivity depends on the density  $g$  evaluated at the firm's surplus. Where  $g$  is high, many firms are near the threshold, and a small change in the gap tips many across it. Where  $g$  is low, few firms are near the threshold, and the same change has little effect.*
- (iv) *If  $G$  has bounded support  $[0, \bar{\xi}]$ , then  $\Lambda(x) = 1$  for  $|x| \geq \sqrt{w\bar{\xi}/B}$ : when the gap is sufficiently large, the surplus exceeds even the highest cost draw, and the firm reprices with certainty. This defines the “inaction band” beyond which all firms adjust. If  $G$  has unbounded support (e.g., exponential),  $\Lambda(x) < 1$  for all finite  $x$  and  $\Lambda(x) \rightarrow 1$  only as  $|x| \rightarrow \infty$ .*

Together, these properties paint a simple picture of the repricing decision. A firm that is well priced ( $x \approx 0$ ) has little to gain from repricing and almost never does. As its price drifts further from the optimum, the gain from repricing grows and exceeds more cost draws, so the repricing probability rises smoothly. Eventually, for a sufficiently large gap, the gain exceeds every possible cost draw, and the firm reprices with certainty. The shape of this transition — how quickly the probability rises from zero to one — is governed entirely by the density  $g$  of menu costs. A dense region of  $g$  near a given surplus level means many firms are close to indifferent there, producing a steep slope in  $\Lambda$ : a small change in the gap tips many across the threshold. A sparse region produces a gentle transition. This is the microeconomic mechanism through which  $G$  shapes aggregate price dynamics.

**Microfoundation of the hazard elasticity** In the power-law family  $G(\xi) = (\xi/\bar{\xi})^\gamma$ , the hazard takes the closed form  $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ , with local exponent  $\nu = 2\gamma$ . This identity *microfounds* the reduced-form hazard elasticity that [Alvarez, Lippi, and Oskolkov \(2022\)](#) take as a sufficient statistic for non-neutrality:  $\nu$  decomposes as *(surplus curvature)*  $\times$  *(cost*

*concentration*), with the 2 coming from the quadratic surplus  $S(x) = Bx^2$  and the  $\gamma$  coming from the cost-distribution shape. The duality therefore supplies the economic anatomy of ALO's reduced-form exponent rather than competing with it as a measurement device: their sufficient statistic summarises non-neutrality without requiring its decomposition; here,  $\nu = 2\gamma$  identifies the two structural primitives that generate it. This is the pricing instantiation of the more general statement of the Remark after Theorem 1: the duality generates the full hazard  $\Lambda(x) = G(S(x)/w)$  from  $(G, S)$ , and the local exponent  $\nu$  is one reading of that generated object.

### 3.4 Stationary distribution of price gaps

To close the model, the cross-sectional density  $f_{ss}$  of price gaps must be characterised. Under trend inflation  $\pi > 0$ , a non-adjusting firm's gap falls by  $\pi$  each period, while an adjusting firm — a firm that reprices with probability  $\Lambda(x)$  — resets to the target  $x^*$ . The cross-section is thus renewed by resets and stretched by the deterministic drift.

**Proposition 7** (Stationary distribution).

*Under  $\Lambda(x) = G(Bx^2/w)$  and trend inflation  $\pi > 0$ , the stationary gap density satisfies the renewal fixed point*

$$f_{ss}(x) = \underbrace{\bar{\Lambda} r(x)}_{\text{reset flow}} + \underbrace{(1 - \Lambda(x + \pi)) f_{ss}(x + \pi)}_{\text{drift flow}}, \quad \bar{\Lambda} = \int \Lambda(x) f_{ss}(x) dx, \quad (18)$$

where  $r$  is the post-reset gap distribution — degenerate at the reset point absent idiosyncratic shocks, smoothed by them otherwise. Equivalently,  $f_{ss}(x)$  is proportional to the probability that a firm has not repriced since its last reset, so it decays along the drift direction at a rate governed by the hazard. Under positive trend inflation  $f_{ss}$  is asymmetric, supported on the inaction region between the reset point and the adjustment boundary. The proof is in Online Appendix C.

The shape of  $G$  governs the cross-section: a concentrated  $G$  (steep hazard) clears firms soon after their gap becomes sizeable, yielding a tight distribution near the reset point, while a dispersed  $G$  (flat hazard) lets firms drift far before repricing, yielding a wide distribution. With the hazard and the stationary distribution in hand, the next question is how the composition of repricing firms shapes aggregate outcomes.

### 3.5 The selection effect

Under state-dependent pricing, firms with the largest price gaps are most likely to reprice, amplifying the aggregate price response relative to Calvo (Golosov and Lucas, 2007). The duality provides a precise characterisation of how  $G$  governs this selection. I first define a measure of selection intensity, then show how it depends on  $G$ .

**Definition 4** (Selection intensity).

*Given a cross-sectional density  $f(x)$  of price gaps, the average absolute adjusted price change is defined as*

$$\overline{|x|}^{adj} := \frac{\int |x| \cdot \Lambda(x) \cdot f(x) dx}{\int \Lambda(x) \cdot f(x) dx} = \frac{E[\Lambda(x) \cdot |x|]}{E[\Lambda(x)]}. \quad (19)$$

*This is the mean absolute price change among repricing firms. The selection intensity is the proportional excess of the average adjuster's absolute price change over the population average:*

$$\mathcal{S} \equiv \frac{\overline{|x|}^{adj}}{E[|x|]} - 1. \quad (20)$$

When  $\Lambda$  is constant (Calvo),  $\mathcal{S} = 0$ : every firm is equally likely to reprice, so adjusters look just like the population. When  $\Lambda$  is a step function (Golosov-Lucas),  $\mathcal{S}$  is at its polar value within the power law family at fixed  $\bar{\Lambda}$ : only firms at the boundary of the inaction band reprice, so every adjuster has a large gap. In between, selection intensity is governed by how sharply  $\Lambda$  increases with  $|x|$ . The next proposition makes the link to  $G$  precise: since  $\Lambda(x) = G(Bx^2/w)$ , the shape of the cost distribution directly determines how strong selection is.

**Proposition 8** (Selection and the menu cost distribution).

*Fix the cross-sectional density  $f$ . The adjustment-weighted mean  $\overline{|x|}^{adj}$  admits the decomposition*

$$\overline{|x|}^{adj} = E[|x|] + \frac{\text{Cov}(\Lambda(x), |x|)}{\bar{\Lambda}},$$

*so that selection intensity  $\mathcal{S}$  has the same sign as  $\text{Cov}(\Lambda(x), |x|)$ . As a consequence:*

- (i)  $\mathcal{S} = 0$  in the Calvo limit (constant  $\Lambda$ ), since the covariance vanishes.
- (ii)  $\mathcal{S} > 0$  whenever  $\Lambda$  is strictly increasing in  $|x|$ . The size of  $\mathcal{S}$  scales with the local elasticity  $\partial \ln \Lambda / \partial \ln |x| = 2(g/G)(Bx^2/w) \cdot (Bx^2/w)$ , i.e., twice the hazard rate of  $G$  at the firm's surplus.
- (iii) In the Golosov-Lucas limit,  $\Lambda$  is a step function and adjusters are concentrated at the inaction-band boundary; this is the polar case of the spectrum, with  $\mathcal{S}$  the largest

attained within the power law family at fixed  $\bar{\Lambda}$ . A general extremality statement across all hazards calibrated to the same average frequency would require a rearrangement-style argument not pursued here.

The proof is given in Online Appendix C.

*Proof.* See Online Appendix C. ■

A concentrated  $G$  (high density near  $\bar{\xi}$ ) means most firms face large menu costs. Only firms with very large gaps find it worthwhile to reprice, so the hazard is steep and selection is strong. A dispersed  $G$  (density spread across  $[0, \bar{\xi}]$ ) means many firms face small costs. Even firms with small gaps reprice frequently, the hazard is flat, and selection is weak. The density  $g$  at the threshold is the key: it determines how many additional firms are tipped into adjustment by a small increase in the gap.

### 3.6 The Calvo–Goloso–Lucas spectrum

The results so far hold for any  $G$  with a positive density. To make the framework quantitatively useful, one needs a tractable parametric family that can span the range of pricing behaviour observed in the data. The two canonical models of price rigidity in the New Keynesian literature — Calvo and Goloso–Lucas — correspond to polar extreme shapes of the hazard  $\Lambda(x)$ . The goal of this subsection is to connect them through a single continuous family indexed by one parameter.

**Calvo (1983)** In the Calvo (1983) model, each firm reprices with a fixed probability  $\lambda \in (0, 1)$  regardless of its price gap. Since  $\Lambda(x) = G(Bx^2/w)$  is always increasing in  $|x|$ , a strictly constant hazard cannot arise from any non-degenerate absolutely continuous  $G$ . The framework therefore does not literally nest Calvo; the following proposition makes the limiting sense precise.

**Proposition 9** (Calvo approximation).

*For the power law family  $G(\xi) = (\xi/\bar{\xi})^\gamma$  with  $\bar{\xi}(\gamma)$  calibrated to maintain a fixed average frequency  $\lambda$ , and assuming  $f_{ss}$  is supported in the interior of the inaction band, the selection wedge  $\kappa - \lambda = 2\gamma\lambda$  vanishes as  $\gamma \rightarrow 0^+$ . In this limit:*

- (i) *The Phillips curve slope converges to the Calvo slope:  $\kappa \rightarrow \lambda$ .*
- (ii) *Non-neutrality converges to the Calvo level:  $1 - \kappa \rightarrow 1 - \lambda$ .*



The hazard  $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$  does not converge to a constant as  $\gamma \rightarrow 0^+$  with fixed  $\bar{\xi}$ : it converges to near-certain adjustment for any  $x \neq 0$  (since  $a^\gamma \rightarrow 1$  for any  $a > 0$ ). The Calvo approximation arises only through the frequency-calibrated limit, in which  $\bar{\xi} \rightarrow \infty$  as  $\gamma \rightarrow 0^+$  so that the hazard remains small and approximately constant over the support of  $f_{ss}$  (the residual variation goes as  $\gamma \log(Bx^2/(w\bar{\xi}))$ , vanishing in the limit). In this regime selection contributes  $2\gamma\lambda$  to the Phillips curve slope, vanishing relative to the frequency component  $\lambda$ .

*Proof.* See Online Appendix C. ■

**Golosov and Lucas (2007): step-function hazard** When  $G$  is degenerate at  $\bar{\xi}$ ,  $\Lambda(x) = \mathbf{1}[Bx^2 \geq w\bar{\xi}]$  — the canonical  $Ss$  rule with inaction band  $\bar{x} = \sqrt{w\bar{\xi}/B}$ . This is the polar case of the spectrum: the step-function hazard generates the largest selection within the power law family at fixed  $\bar{\Lambda}$  and dramatically reduces non-neutrality relative to Calvo.

**The power law family** The power law  $G(\xi) = (\xi/\bar{\xi})^\gamma$ ,  $\gamma > 0$ , provides a tractable one-parameter family that spans a selection spectrum whose low- $\gamma$  limit is Calvo-equivalent in the frequency-calibrated sense (Proposition 9) and whose high- $\gamma$  limit is Golosov-Lucas. The parameter  $\gamma$  is the degree of cost concentration. When  $\gamma$  is low, the density  $g$  is decreasing: most firms draw small menu costs, adjustment is cheap, and the hazard rises gently with the gap — close to Calvo. When  $\gamma$  is high, the density concentrates near  $\bar{\xi}$ : most firms draw large menu costs, only firms with very large gaps find it worthwhile to reprice, and the hazard rises steeply — close to Golosov-Lucas. The induced hazard and cost function are:

$$\Lambda(x) = \left( \frac{Bx^2}{w\bar{\xi}} \right)^\gamma, \quad C(\varphi) = \frac{\gamma}{\gamma+1} w\bar{\xi} \varphi^{1+1/\gamma}. \quad (21)$$

The hazard expression applies for  $|x| \leq \sqrt{w\bar{\xi}/B}$ ; for larger gaps  $\Lambda(x) = 1$ , defining the inaction band boundary.  $\Lambda$  is continuous but not  $C^1$  at  $|x| = \sqrt{w\bar{\xi}/B}$ ; boundary sets have measure zero under absolutely continuous  $f_{ss}$ . This family nests the uniform distribution ( $\gamma = 1$ ) and extends to right-skewed cost distributions ( $\gamma < 1$ ).

**Example 1** (Power law menu costs).

For  $G(\xi) = (\xi/\bar{\xi})^\gamma$ , the following relations hold:

- (i)  $G^{-1}(\varphi) = \bar{\xi} \varphi^{1/\gamma}$ .
- (ii)  $C(\varphi) = \frac{\gamma}{\gamma+1} w\bar{\xi} \varphi^{(\gamma+1)/\gamma}$ .
- (iii)  $C'(\varphi) = w\bar{\xi} \varphi^{1/\gamma}$ ,  $C''(\varphi) = \frac{w\bar{\xi}}{\gamma} \varphi^{(1-\gamma)/\gamma}$ .

$$(iv) \quad \Lambda(x) = \left( \frac{Bx^2}{w\xi} \right)^\gamma.$$

The parameter  $\gamma$  governs the entire spectrum:  $\gamma = 1$  gives uniform costs and quadratic  $C$  (Dotsey, King, and Wolman, 1999; Khan and Thomas, 2008);  $\gamma \rightarrow \infty$  gives the step-function hazard of Golosov and Lucas (2007);  $\gamma < 1$  gives right-skewed costs with a concave hazard and weak selection;  $\gamma \rightarrow 0$  gives the “cheap adjustment” limit where nearly all firms reprice. Note that the  $\gamma \rightarrow 0$  limit with fixed  $\bar{\xi}$  yields near-certain adjustment, not a constant interior hazard. The Calvo approximation requires the frequency-calibrated limit of Proposition 9, which adjusts  $\bar{\xi}$  with  $\gamma$  to hold  $\bar{\Lambda}$  fixed.

### 3.7 Aggregate inflation and the Phillips curve

The repricing hazard  $\Lambda(x)$  governs individual pricing behaviour. Because the duality gives  $\Lambda(x) = G(Bx^2/w)$  in closed form, aggregate inflation can be written directly in terms of  $G$  — enabling analytical decompositions that would require numerical computation without the duality. Let  $f_t(x)$  denote the cross-sectional density of price gaps at time  $t$ . Aggregate inflation is

$$\pi_t = E_{f_t}[\Lambda(x_t) \cdot x_t] = \int \Lambda(x) x f_t(x) dx. \quad (22)$$

Each firm contributes in proportion to  $\Lambda(x) \cdot x$ . Firms with large gaps and high repricing probabilities contribute the most; firms near the optimum contribute almost nothing.

To understand how aggregate inflation responds to nominal disturbances, consider a small aggregate shock  $\mu_t$  that shifts the desired price of all firms by  $\mu_t$ , moving the cross-sectional distribution of gaps from the steady-state  $f_{ss}(x)$  to  $f_{ss}(x - \mu_t)$ .

**Proposition 10** (Nonlinear Phillips curve).

*Under the quadratic surplus approximation  $S(x) = Bx^2$  and menu cost distribution  $G$ , aggregate inflation satisfies*

$$\pi_t = E_t \left[ G \left( \frac{Bx_t^2}{w} \right) \cdot x_t \right]. \quad (23)$$

*For small aggregate disturbances around the steady state with gap distribution  $f_{ss}$ :*

$$\pi_t \approx \kappa(G) \cdot \mu_t, \quad (24)$$

*where the distribution-dependent slope is*

$$\kappa(G) := \underbrace{\int G \left( \frac{Bx^2}{w} \right) f_{ss}(x) dx}_{\text{extensive margin (frequency)}} + \underbrace{\int g \left( \frac{Bx^2}{w} \right) \frac{2Bx^2}{w} f_{ss}(x) dx}_{\text{intensive margin (selection)}}. \quad (25)$$

*Proof.*

Write  $f_t(x) = f_{ss}(x - \mu_t)$  and differentiate  $\pi_t = \int G(Bx^2/w) x f_{ss}(x - \mu_t) dx$  with respect to  $\mu_t$  at  $\mu_t = 0$ , using  $\partial_\mu f_{ss}(x - \mu) = -\partial_x f_{ss}(x - \mu)$ . Integrating by parts (assuming boundary terms vanish) and expanding  $\frac{d}{dx}[G(Bx^2/w) \cdot x]$  yields the two terms: the first from  $G$  itself (extensive margin) and the second from  $g \cdot 2Bx^2/w$  (intensive margin). The full derivation is in Online Appendix C.  $\blacksquare$

The extensive margin (first term) captures the average repricing frequency — the only margin under Calvo. The intensive margin (second term) captures selection: a nominal shock tips firms near the threshold into adjustment, amplifying the price response. It is absent under Calvo and dominant under Golosov-Lucas. The slope  $\kappa(G)$  is derived under a partial-equilibrium one-time translation of  $f_{ss}$ . General-equilibrium feedbacks — strategic complementarities, endogenous wages (which enter the hazard via  $\Lambda(x) = G(Bx^2/w)$ ), and post-shock evolution of  $f_t$  that is not a pure translation of  $f_{ss}$  — modify the quantitative slope; the present results characterise the partial-equilibrium component, complementing the general-equilibrium treatment in [Alvarez, Le Bihan, and Lippi \(2016\)](#).

The slope  $\kappa(G)$  varies across the spectrum because  $G$  governs the relative strength of these two margins. The following corollary traces out the implications for the canonical cases.

**Corollary 4** (How  $G$  governs the Phillips curve).

*Specialising the slope  $\kappa(G)$  in (25) to the canonical menu-cost distributions shows how the balance between the two margins shifts as  $G$  concentrates: dispersed  $G$  leaves only the frequency term, while concentrated  $G$  makes selection dominate. The three benchmark cases are:*

- (i) **Calvo limit:** *When  $G$  is highly dispersed,  $g$  is small across the region where  $f_{ss}$  has mass, and the intensive margin is negligible. The slope approaches  $\kappa \approx \bar{\Lambda}$ , approximating the textbook New Keynesian Phillips Curve.*
- (ii) **Golosov-Lucas:** *When  $G$  is degenerate,  $G(Bx^2/w) = \mathbf{1}[|x| \geq \bar{x}]$  and  $g = \delta_{\bar{x}}$ . The intensive margin dominates: inflation is driven by selection of firms at the threshold. The slope is very steep, implying low non-neutrality.*
- (iii) **Uniform:** *When  $G = U[0, \bar{\xi}]$ , the slope simplifies to*

$$\kappa^U = \frac{3B}{w\bar{\xi}} E_{ss}[x^2], \quad (26)$$

the second moment of the gap distribution, which reduces to  $\frac{3B}{w\xi} \text{Var}_{ss}(x)$  when gaps are mean-zero (the zero-trend-inflation benchmark) and otherwise also reflects the mean drift. The slope depends on the steady-state dispersion of price gaps. Economies with more dispersed gaps have steeper Phillips curves, because the selection margin is stronger when more firms have large gaps near the threshold.

(iv) **Power law  $\gamma$ :** The slope  $\kappa$  is strictly increasing in  $\gamma$ . Higher  $\gamma$  means more concentrated menu costs, a steeper hazard, stronger selection, and a steeper Phillips curve.

### 3.8 Non-neutrality decomposition

The degree of monetary non-neutrality is inversely related to the price flexibility index  $\kappa$ . When  $\kappa$  is high, prices adjust quickly and real effects are small. When  $\kappa$  is low, prices are sticky and real effects are large. Because the duality expresses both  $\kappa$  and  $\bar{\Lambda}$  as explicit functions of  $G$  and  $g$ , it yields a clean decomposition of  $\kappa$  into two forces. For the power law family, this decomposition is available in closed form.

**Proposition 11** (Frequency vs. selection).

The on-impact real effect of a monetary shock  $dm$  is  $(1 - \kappa(G)) dm$ , well-defined provided  $\kappa(G) \leq 1$ ; more generally, cumulative output is inversely related to  $\kappa(G)$  (see Table 2 note for the  $\kappa > 1$  regime). The flexibility index decomposes as

$$\kappa(G) = \underbrace{\bar{\Lambda}}_{\text{frequency}} + \underbrace{(\kappa - \bar{\Lambda})}_{\text{selection wedge}}, \quad (27)$$

where  $\bar{\Lambda} = E_{f_{ss}}[\Lambda(x)]$  is the average adjustment frequency. Non-neutrality is governed by the selection wedge  $\kappa - \bar{\Lambda}$ .

*Proof.* The decomposition is the algebraic identity  $\kappa = \bar{\Lambda} + (\kappa - \bar{\Lambda})$ ; the two terms are the extensive- and intensive-margin integrals of Proposition 10, equation (25). ■

The decomposition separates two conceptually distinct channels. Frequency ( $\bar{\Lambda}$ ) measures how often firms reprice. The selection wedge ( $\kappa - \bar{\Lambda}$ ) measures how well those repricings are targeted at the firms that need them most. Under Calvo,  $\kappa = \bar{\Lambda} = \lambda$  and selection contributes nothing: repricing is random. For any absolutely continuous  $G$ , provided  $f_{ss}$  has non-degenerate mass on the interior region where  $\Lambda$  is strictly increasing, the selection wedge is strictly positive: firms with larger gaps are more likely to reprice, amplifying the aggregate price response beyond what frequency alone would produce.

This distinction has a striking implication. Two economies can share the same repricing frequency yet differ dramatically in non-neutrality. The difference is entirely attributable to the selection wedge, which depends on the shape of  $G$ .

**Corollary 5** (Ordering of non-neutrality within the power law family).

*For the power law family  $G(\xi) = (\xi/\bar{\xi})^\gamma$  with  $\bar{\xi}(\gamma)$  calibrated to a fixed average frequency  $\bar{\Lambda}$ , and assuming  $f_{ss}$  is supported in the interior of the inaction band:*

- (i) **Calvo benchmark** ( $\gamma \rightarrow 0^+$ ): *selection wedge = 0; largest non-neutrality.*
- (ii) **Interior**  $\gamma > 0$ : *selection wedge =  $2\gamma\bar{\Lambda} > 0$ , strictly increasing in  $\gamma$  (Proposition 12); intermediate non-neutrality.*
- (iii) **Golosov-Lucas** ( $\gamma \rightarrow \infty$ ): *selection wedge attains its supremum within the power law family; smallest non-neutrality.*

*The interior-support hypothesis is approximate under typical normal-shock calibrations; under that approximation the ordering above holds to the same approximation. A general rearrangement-style ordering across all absolutely continuous  $G$  requires additional regularity beyond the present framework and is not pursued here.*

Within the power law family, the ordering is exact:  $\gamma$  alone determines selection, non-neutrality, and the Phillips curve slope.

**Proposition 12** (Monotonicity in  $\gamma$ ).

*For the power law family  $G(\xi) = (\xi/\bar{\xi})^\gamma$  with  $B/w$  fixed and  $\bar{\xi}(\gamma)$  chosen so that the average adjustment frequency is constant across  $\gamma$ , and assuming  $f_{ss}$  is supported in the interior of the inaction band  $|x| < \sqrt{w\bar{\xi}/B}$ :*

- (i) *The normalised selection wedge  $(\kappa - \bar{\Lambda})/\bar{\Lambda} = 2\gamma$  is strictly increasing in  $\gamma$ .*
- (ii) *Monetary non-neutrality is strictly decreasing in  $\gamma$ .*
- (iii) *The Phillips curve slope at  $\mu = 0$  is strictly increasing in  $\gamma$ .*

*The interior-support hypothesis holds approximately under typical normal-shock calibrations, where the inaction band is wide relative to the shock variance; in that regime the bullets above hold to the same approximation. Thus  $\gamma$  indexes “how Calvo-like” the pricing environment is. Low  $\gamma$  (dispersed costs, weak selection, large non-neutrality) resembles Calvo. High  $\gamma$  (concentrated costs, strong selection, small non-neutrality) resembles Golosov-Lucas. The distribution  $G$  determines the economy’s position on this spectrum.*

*Proof.* See Online Appendix C. ■

A natural question is whether repricing frequency alone determines non-neutrality. The answer is no. Frequency is only half the story; what matters equally is how that repricing is allocated across firms. The following proposition makes this precise.

**Proposition 13** (Frequency does not determine non-neutrality).

*Fix  $f_{ss}$  and  $B/w$ . There exist absolutely continuous distributions  $G_1, G_2$  with:*

- (i) *The same average repricing frequency:  $\bar{\Lambda}_1 = E_{f_{ss}}[\Lambda_1(x)] = E_{f_{ss}}[\Lambda_2(x)] = \bar{\Lambda}_2$ .*
- (ii) *Different flexibility indices:  $\kappa(G_1) \neq \kappa(G_2)$ , hence different selection wedges and different real effects of nominal shocks.*

*Proof.* See Online Appendix C. ■

For the power law family, the decomposition takes a particularly clean form. The ratio of total flexibility to frequency is pinned down by  $\gamma$  alone — independent of the steady-state distribution, the surplus parameter, or the wage.

**Corollary 6** (Selection-frequency ratio for power laws).

*For  $G(\xi) = (\xi/\bar{\xi})^\gamma$ , provided the tail mass of  $f_{ss}$  outside the inaction band is negligible (exact when  $f_{ss}$  is supported in the interior  $|x| < \sqrt{w\xi/B}$ ):*

$$\kappa = (1 + 2\gamma) \bar{\Lambda}, \quad \text{selection wedge} = 2\gamma \bar{\Lambda}. \quad (28)$$

*The ratio  $\kappa/\bar{\Lambda} = 1 + 2\gamma$  depends only on  $\gamma$ , not on  $f_{ss}$ ,  $B$ , or  $w$ .*

*Proof.*

For  $G(\xi) = (\xi/\bar{\xi})^\gamma$ , the density is  $g(\xi) = (\gamma/\bar{\xi})(\xi/\bar{\xi})^{\gamma-1}$ . In the interior of the band,  $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma < 1$ , and evaluating the intensive-margin integrand at  $\xi = Bx^2/w$ :

$$g\left(\frac{Bx^2}{w}\right) \frac{2Bx^2}{w} = \frac{\gamma}{\bar{\xi}} \left(\frac{Bx^2}{w\bar{\xi}}\right)^{\gamma-1} \frac{2Bx^2}{w} = 2\gamma \left(\frac{Bx^2}{w\bar{\xi}}\right)^\gamma = 2\gamma \Lambda(x).$$

Substituting into (25):  $\kappa = \int \Lambda(x) f_{ss}(x) dx + \int 2\gamma \Lambda(x) f_{ss}(x) dx = (1 + 2\gamma) \bar{\Lambda}$ . The identity requires that the uncapped power law expression applies  $f_{ss}$ -a.s., i.e., that  $f_{ss}$  has negligible mass outside the inaction band. In the calibrated model with normal shocks this condition is satisfied: the band is wide relative to the shock variance. ■

Table 2: Same repricing frequency, different selection wedges

| Distribution $G$                     | $\bar{\Lambda}$ | $\kappa$ | $\kappa - \bar{\Lambda}$ | $1 - \kappa$ |
|--------------------------------------|-----------------|----------|--------------------------|--------------|
| Calvo                                | 0.10            | 0.10     | 0                        | 0.90         |
| Power law ( $\gamma = \frac{1}{2}$ ) | 0.10            | 0.20     | 0.10                     | 0.80         |
| Exponential                          | 0.10            | 0.27     | 0.17                     | 0.73         |
| Uniform ( $\gamma = 1$ )             | 0.10            | 0.30     | 0.20                     | 0.70         |
| Power law ( $\gamma = 2$ )           | 0.10            | 0.50     | 0.40                     | 0.50         |
| Power law ( $\gamma = 3$ )           | 0.10            | 0.70     | 0.60                     | 0.30         |

*Notes:* Each row is calibrated to  $\bar{\Lambda} = 0.10$ . The last column,  $1 - \kappa$ , is proportional to the cumulative real output effect of a monetary shock. For high  $\gamma$  and high  $\bar{\Lambda}$ ,  $\kappa$  can exceed one; the linearisation  $\pi \approx \kappa\mu$  then overstates the first-period price response, and the non-neutrality measure  $1 - \kappa$  loses its direct interpretation. The formula remains valid as a local slope but should be interpreted with caution when  $\kappa > 1$ . *Calvo entry:* The Calvo row is the  $\gamma \rightarrow 0^+$  limit of the power-law family with  $\bar{\xi}(\gamma) \rightarrow \infty$  to hold  $\bar{\Lambda}$  fixed (Proposition 9). At interior  $\gamma$  the hazard remains state-dependent; what vanishes is the selection wedge, not the state dependence of the individual hazard. *Power law entries:*  $\kappa = (1 + 2\gamma)\bar{\Lambda}$ , exact when  $f_{ss}$  is supported in the interior of the inaction band; approximate when the tail mass outside the band is negligible (the typical case under standard menu cost calibrations). *Exponential entry:*  $G(\xi) = 1 - e^{-\lambda\xi}$  with  $f_{ss} = N(0, \sigma_x^2)$ . Setting  $\rho = B\sigma_x^2/w$ , the  $\chi^2$  moment generating function gives  $\bar{\Lambda} = 1 - (1 + 2\lambda\rho)^{-1/2}$  and intensive margin  $2\lambda\rho(1 + 2\lambda\rho)^{-3/2}$ . Calibrating  $\lambda\rho$  to match  $\bar{\Lambda} = 0.10$  gives  $\kappa \approx 0.27$ .

Table 2 illustrates. Six distributions calibrated to the same repricing frequency  $\bar{\Lambda} = 0.10$  produce selection wedges ranging from 0 to 0.60. The implied cumulative output effects differ by a factor of three.

Figure 4 complements the table with the dynamic counterpart. The four power law economies plotted span from near-Calvo at  $\gamma = 0.20$ , through uniform at  $\gamma = 1$ , to near-Golosov-Lucas at  $\gamma = 5$ ; each is calibrated to the same quarterly repricing frequency. Each economy is hit with a one-time 1% monetary shock. The normalised output gap  $y(t)/\delta$  is tracked as prices catch up to the new money supply.

The mechanism is selection: low  $\gamma$  means nearly random repricing and large persistent output effects; high  $\gamma$  means targeted repricing and small transient effects. The impulse responses are plotted in Figure 4 (Section 4.4).

### 3.9 Summary of the spectrum

Table 3 collects the pricing results: a single object, the menu cost distribution  $G$ , indexes the entire spectrum from Calvo (dispersed  $G$ , vanishing selection wedge) through uniform  $G$  (quadratic costs, linear hazard, moderate selection) to Golosov-Lucas (point mass, step-



function hazard, polar selection within the power law family). The power law  $\gamma$  traces this spectrum continuously; the empirical question — where real economies sit — is taken up in Section 4.

Table 3: The state-dependence spectrum

|                     | <b>Calvo</b>         | <b>Uniform <math>G</math></b> | <b>Golosov-Lucas</b>      |
|---------------------|----------------------|-------------------------------|---------------------------|
| Distribution $G$    | Dispersed            | $U[0, \bar{\xi}]$             | Point mass at $\bar{\xi}$ |
| $C(\varphi)$        | Highly curved        | Quadratic                     | Linear on $\{0, 1\}$      |
| $C'''(\varphi)$     | $\rightarrow \infty$ | $w\bar{\xi}$                  | 0 or $\infty$             |
| Hazard $\Lambda(x)$ | $\approx$ constant   | $Bx^2/(w\bar{\xi})$           | Step function             |
| Selection           | $\approx$ none       | Moderate                      | Maximum                   |
| NKPC slope          | $\approx$ constant   | State-dependent               | Very steep                |
| Non-neutrality      | Maximum              | Intermediate                  | Minimum                   |
| Power law $\gamma$  | $\rightarrow 0^+$    | 1                             | $\rightarrow \infty$      |

*Notes:* Three reference points on the power law spectrum  $G(\xi) = (\xi/\bar{\xi})^\gamma$ . The Calvo column is the frequency-calibrated limit  $\gamma \rightarrow 0^+$  with  $\bar{\xi}(\gamma) \rightarrow \infty$  to hold  $\bar{\Lambda}$  fixed (Proposition 9); Uniform  $G$  corresponds to  $\gamma = 1$ ; Golosov-Lucas is the polar limit  $\gamma \rightarrow \infty$  at fixed  $\bar{\xi}$ . The hazard, NKPC slope, and non-neutrality rows describe the aggregate consequences of the cost shape: dispersed  $G$  produces near-random repricing and near-Calvo aggregates; a point mass at  $\bar{\xi}$  produces the  $Ss$ -style step hazard of Golosov and Lucas (2007). The empirical location on this spectrum is taken up in Section 4.

## 4 Quantitative analysis

The theoretical results of Sections 2–3 establish that the distribution  $G$  is the structural primitive governing adjustment dynamics. The closed-form hazard  $\Lambda(x) = G(Bx^2/w)$  suggests an immediate identification route: for the power law family,  $\log \Lambda$  is linear in  $\log |x|$  with slope  $2\gamma$  (equivalently, linear in  $\log(x^2)$  with slope  $\gamma$ ), so  $\gamma$  can in principle be read off from the hazard shape alone, without solving the full model. This section asks: what does  $G$  look like in the data? The answer disciplines the theory and quantifies the frequency-selection decomposition.

The quantitative analysis proceeds in three parts. Section 4.1 describes the hazard-based identification and notes that a model-free direct implementation on standard micro price datasets (scanner or CPI) is infeasible without observing the price gap  $x$  directly; model-based proxies for  $x$  exist but are not pursued here. Section 4.2 therefore calibrates  $\gamma$  by matching CPI-based price-change moments: kurtosis and the fraction of small changes in a heterogeneous- $\sigma_z$  simulation. This produces an illustrative calibrated value, not a fully identified estimate, with the caveats discussed below. A robustness exercise on Dominick’s

scanner data is summarised in Section 4.4 and detailed in Online Appendix. Section 4.5 solves the full dynamic model globally and uses the routine to check the analytical decomposition as a tight approximation across the spectrum and to produce state-dependent impulse responses. The hazard-based route remains a testable prediction of the framework for datasets that permit constructing  $x$  directly.

## 4.1 A testable prediction: hazard-based identification

Before turning to the illustrative moment-matching calibration implemented below, I record one prediction the framework makes for future empirical work. The theoretical hazard for the power law family is  $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ . Taking logs gives

$$\log \Lambda(x) = 2\gamma \log |x| + \text{const.} \quad (29)$$

The slope of  $\log \Lambda$  against  $\log |x|$  is  $2\gamma$  (equivalently, against  $\log(x^2)$  the slope is  $\gamma$ ). This identifies  $\gamma$  from the hazard shape alone, independently of  $\bar{\xi}$ ,  $B$ , and  $w$ . A Monte Carlo exercise (Figure D.1 in the Online Appendix) confirms recovery to mean absolute error 0.012 and  $R^2 > 0.99$ .

Implementing this on actual data requires observing the price gap  $x$  at each decision point, which is not recorded in standard micro price datasets (scanner or CPI) — only price changes  $\Delta p$ . Model-based proxies for  $x$  exist but are not pursued here. Datasets with posted reference prices or shelf tags permit constructing  $x$  directly; that is the natural setting for the hazard-based test. The remainder of this section uses moment matching to obtain an illustrative calibrated value of  $\gamma$ , not a fully identified estimate.

## 4.2 Calibration and moment matching

The baseline calibration sets  $B = w = 1$  (normalisation) and  $\sigma_z = 0.04$  (quarterly standard deviation of idiosyncratic desired-price shocks, consistent with Golosov and Lucas 2007 and Nakamura and Steinsson 2008). For each  $\gamma$ , the upper bound  $\bar{\xi}$  is calibrated to match a quarterly repricing frequency of 24%, corresponding to approximately 8–9% monthly excluding temporary sales (Nakamura and Steinsson, 2008); under constant monthly hazards,  $1 - (1 - 0.085)^3 \approx 0.234$ . Throughout,  $\varphi$  is interpreted as a per-period probability.

In the baseline model with homogeneous  $\sigma_z$ , simulated kurtosis ranges from 1.33 (at  $\gamma = 2$ ) to 2.79 (at  $\gamma = 0.15$ ) across the  $\gamma$  grid (Table 4, left panel) — well below the empirical kurtosis of 4–6, a well-known limitation of single-product menu cost models with a homogeneous shock process. Following Midrigan (2011), the extended model draws per-

product  $\sigma_z$  from a lognormal distribution with mean 0.04 and coefficient of variation  $CV = 1$ . The CV is imposed, not estimated; this generates kurtosis in the empirical range while preserving the identification of  $\gamma$  from the hazard shape. Sensitivity to  $CV \in \{0.5, 1.5\}$  is left for future quantitative work.

Table 4: Model moments: baseline and heterogeneous  $\sigma_z$

| $\gamma$     | Baseline (homogeneous $\sigma_z$ ) |                 |        |     |          | Heterogeneous $\sigma_z$ (CV = 1) |                 |        |     |          |
|--------------|------------------------------------|-----------------|--------|-----|----------|-----------------------------------|-----------------|--------|-----|----------|
|              | Freq                               | $E[ \Delta p ]$ | Kurt   | Sm% | $\kappa$ | Freq                              | $E[ \Delta p ]$ | Kurt   | Sm% | $\kappa$ |
| 0.15         | 0.24                               | 7.2%            | 2.79   | 17% | 0.31     | 0.24                              | 7.9%            | 7.38   | 28% | 0.31     |
| 0.25         | 0.24                               | 7.5%            | 2.47   | 14% | 0.36     | 0.24                              | 8.8%            | 6.18   | 21% | 0.36     |
| 0.40         | 0.24                               | 7.8%            | 2.11   | 10% | 0.43     | 0.24                              | 9.7%            | 4.86   | 14% | 0.44     |
| 0.60         | 0.24                               | 8.1%            | 1.80   | 6%  | 0.53     | 0.24                              | 10.3%           | 4.49   | 9%  | 0.53     |
| 0.80         | 0.24                               | 8.3%            | 1.61   | 4%  | 0.63     | 0.24                              | 10.6%           | 4.44   | 5%  | 0.63     |
| 1.00         | 0.24                               | 8.4%            | 1.51   | 2%  | 0.72     | 0.24                              | 10.8%           | 4.37   | 3%  | 0.73     |
| 1.50         | 0.24                               | 8.6%            | 1.38   | 1%  | 0.96     | 0.24                              | 11.0%           | 4.31   | 1%  | 0.97     |
| 2.00         | 0.24                               | 8.7%            | 1.33   | 0%  | 1.20     | 0.24                              | 11.1%           | 4.28   | 0%  | 1.21     |
| Data targets | 8–11%                              | 4–6             | 25–40% |     |          | 8–11%                             | 4–6             | 25–40% |     |          |

*Notes:* All economies calibrated to quarterly repricing frequency  $\bar{\Lambda} = 0.24$ . “Freq” is the simulated frequency (confirming calibration). “ $E[|\Delta p|]$ ” is the mean absolute (non-sale) price change. “Kurt” is the raw kurtosis. “Sm%” is the fraction of price changes smaller than 2.5% in absolute value.  $\kappa = (1+2\gamma)\bar{\Lambda}$  holds exactly when  $f_{ss}$  is supported in the interior of the inaction band and approximately to the extent that tail mass falls outside it; in the calibrated economies the approximation is tight under the standard normal-shock specification. Data targets from [Nakamura and Steinsson \(2008\)](#), [Midrigan \(2011\)](#), and [Alvarez, Le Bihan, and Lippi \(2016\)](#). 15,000 firms, 150 quarters per simulation.

### 4.3 Calibration of $\gamma$ : hazard-exponent and kurtosis routes

**Benchmark calibrated value (hazard-exponent route)** The benchmark calibration anchors  $\gamma$  to the heterogeneity-corrected hazard exponent recovered by [Alvarez, Lippi, and Oskolkov \(2022\)](#) via the structural mapping  $\nu = 2\gamma$  of the Remark after Theorem 1. ALO’s Kotlarski-style deconvolution of scraped CPI data ([Cavallo, 2018](#)) delivers a roughly quadratic hazard  $\Lambda(x) \propto |x|^\nu$  with  $\nu \approx 2$ . The structural reading inside this framework is

$$\hat{\gamma} = \nu/2 \approx 1. \quad (30)$$

This is the empirically preferred value because ALO’s correction strips out the unobserved cross-product heterogeneity that inflates raw price-change kurtosis; their recovered hazard

exponent is therefore the cleanest available identification of the structural cost-shape parameter in the generalised-hazard family.

**Alternative (kurtosis-matching) calibration** A purely moment-matching alternative is also available: matching the raw empirical kurtosis  $\approx 5.0$  in the heterogeneous- $\sigma_z$  model of Section 4.2 yields  $\hat{\gamma} \approx 0.4$ , with a selection share of about 44%. This is reported alongside the benchmark for transparency, but the structural hazard-exponent identification is the one that anchors the headline.

**Implied decomposition at  $\hat{\gamma} = 1$**  The flexibility index is

$$\kappa = (1 + 2 \times 1) \times 0.24 = 0.72.$$

The frequency component is  $\bar{\Lambda} = 0.24$ , and the selection wedge is  $\kappa - \bar{\Lambda} = 0.48$ . Selection contributes  $\frac{2}{3}$  of  $\kappa$ :

$$\frac{\kappa - \bar{\Lambda}}{\kappa} = \frac{0.48}{0.72} = \frac{2}{3} \approx 67\%.$$

*Selection is the dominant component of monetary non-neutrality.* Non-neutrality is proportional to  $1 - \kappa = 0.28$ . Relative to the same-frequency Calvo benchmark ( $1 - \bar{\Lambda} = 0.76$ ), this is  $(1 - \kappa)/(1 - \bar{\Lambda}) = 0.37$ : selection reduces non-neutrality by roughly 63% relative to Calvo at the same frequency. Table 5 summarises and contrasts with the kurtosis-matching alternative.

Table 5: Frequency-selection decomposition: benchmark vs. kurtosis-matching

|                                              | Benchmark ( $\hat{\gamma} = 1$ ) |                   | Kurtosis-matching ( $\hat{\gamma} = 0.4$ ) |                   |
|----------------------------------------------|----------------------------------|-------------------|--------------------------------------------|-------------------|
|                                              | Value                            | Share of $\kappa$ | Value                                      | Share of $\kappa$ |
| Frequency ( $\bar{\Lambda}$ )                | 0.24                             | 33%               | 0.24                                       | 56%               |
| Selection wedge ( $\kappa - \bar{\Lambda}$ ) | 0.48                             | 67%               | 0.19                                       | 44%               |
| Flexibility ( $\kappa$ )                     | 0.72                             | 100%              | 0.43                                       | 100%              |
| Non-neutrality ( $1 - \kappa$ )              | 0.28                             | —                 | 0.57                                       | —                 |

*Notes:*  $\kappa = (1 + 2\hat{\gamma})\bar{\Lambda}$ . Benchmark:  $\hat{\gamma} = 1$  from ALO's hazard-exponent route ( $\nu = 2\gamma$ ,  $\nu \approx 2$ ); selection share =  $2\gamma/(1 + 2\gamma) = \frac{2}{3}$ . Kurtosis-matching alternative:  $\hat{\gamma} = 0.4$  from heterogeneous- $\sigma_z$  kurtosis. Non-neutrality as a fraction of Calvo at the same frequency:  $(1 - \kappa)/(1 - \bar{\Lambda}) = 0.28/0.76 \approx 0.37$  at  $\hat{\gamma} = 1$  vs.  $0.57/0.76 \approx 0.75$  at  $\hat{\gamma} = 0.4$ .

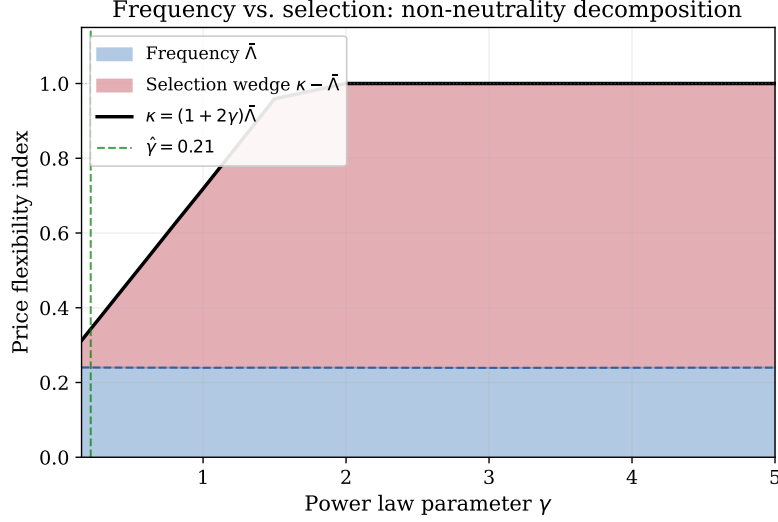


Figure 1: Frequency-selection decomposition of  $\kappa$  across the spectrum

*Notes:* Decomposition of the Phillips curve slope  $\kappa = \bar{\Lambda} + (\kappa - \bar{\Lambda})$  as a function of  $\gamma$  within the power-law family, holding  $\bar{\Lambda} = 0.24$  fixed. The frequency component is constant by construction; the selection wedge  $2\gamma\bar{\Lambda}$  rises linearly in  $\gamma$ . The vertical dashed line marks the benchmark  $\hat{\gamma} = 1$  from the hazard-exponent route ( $\nu = 2\gamma$ ,  $\nu \approx 2$ ), where selection contributes  $\frac{2}{3}$  of  $\kappa$ .

**Sensitivity of the selection share** Since the selection share  $2\gamma/(1 + 2\gamma)$  is an analytic function of  $\gamma$ , its sensitivity is transparent. The benchmark  $\hat{\gamma} = 1$  delivers  $\frac{2}{3} \approx 67\%$ . The kurtosis-matching alternative  $\hat{\gamma} = 0.4$  delivers 44%. The full range across the literature-relevant interval  $\gamma \in [0.15, 1.5]$  spans selection shares from 23% to 75%, monotonically increasing in  $\gamma$ . The formula  $\kappa = (1 + 2\gamma)\bar{\Lambda}$ , exact when  $f_{ss}$  is supported in the interior of the inaction band and approximately so otherwise, is independent of  $\bar{\Lambda}$ , so the selection share does not depend on the targeted repricing frequency.

**Comparison with the literature** The benchmark  $\hat{\gamma} = 1$  places the calibration squarely at the Khan-Thomas (Khan and Thomas, 2008) uniform- $G$  point of the spectrum, with selection as the dominant flexibility margin. Alvarez, Lippi, and Oskolkov (2022) report a heterogeneity-corrected price-change kurtosis  $\approx 2$  (against the 4–6 commonly reported in uncorrected data), consistent with the hazard-exponent reading  $\hat{\gamma} = 1$  once one accepts the structural mapping  $\nu = 2\gamma$ . Midrigan (2011)’s multi-product structure that generates many small price changes alongside occasional large ones is consistent with a right-skewed  $G$  ( $\gamma < 1$ , i.e., the kurtosis-matching alternative reading rather than the benchmark). Alvarez, Le Bihan, and Lippi (2016) report a kurtosis-frequency ratio of approximately 3.5; under the present framework the model’s CIR ratio is  $\text{Kurt}/(6 \cdot \text{Freq})$  which varies with the kurtosis

the model produces at the chosen  $\hat{\gamma}$  (see Figure 2). Vavra (2014) documents that the cross-sectional variance of price changes comoves with inflation — a prediction that follows directly from the state-dependent hazard  $\Lambda(x) = G(Bx^2/w)$  whenever  $G$  generates a non-constant hazard.

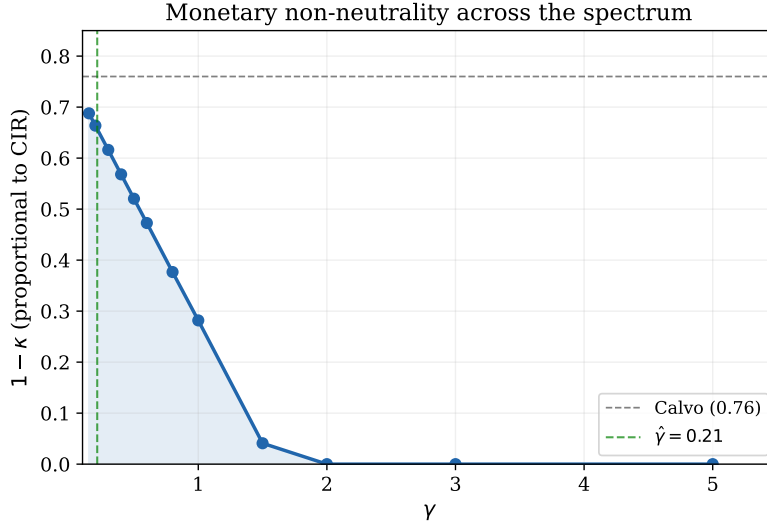


Figure 2: ABLL’s cumulative impulse response (CIR) as a function of  $\gamma$

*Notes:*  $\text{CIR}(\gamma) = \text{Kurt}(\gamma)/(6 \cdot \bar{\Lambda})$  under the power-law family, with  $\bar{\Lambda} = 0.24$  fixed across  $\gamma$ . The full curve maps the entire Calvo–Goloso–Lucas spectrum onto ABLL’s single sufficient statistic. The kurtosis-matching alternative  $\hat{\gamma} = 0.4$  produces CIR  $\approx 3.4$ , close to ABLL’s empirical value of 3.5; the benchmark  $\hat{\gamma} = 1$  from the hazard-exponent route corresponds to a lower CIR consistent with ALO’s heterogeneity-corrected kurtosis  $\approx 2$ .

#### 4.4 Hazard functions and impulse responses

Figure 3 plots the simulated repricing hazard for four values of  $\gamma$ , along with the theoretical prediction (dashed). The hazard is concave in  $|x|$  for  $\gamma < 1$  (firms with moderate gaps adjust frequently, but the marginal increase for very large gaps is small) and convex for  $\gamma > 1$  (adjustment is concentrated sharply at the threshold). The benchmark  $\hat{\gamma} = 1$  sits at the knife-edge between the two regimes, producing a quadratic hazard  $\Lambda(x) \propto x^2$  consistent with the heterogeneity-corrected hazard exponent of Alvarez, Lippi, and Oskolkov (2022); the kurtosis-matching alternative  $\hat{\gamma} \approx 0.4$  produces a gently increasing concave hazard in the spirit of the smooth adjustment functions estimated by Caballero and Engel (1993).

Figure 4 shows the output impulse response to a one-time 1% monetary shock. All economies share the same repricing frequency; they differ only in  $\gamma$ . The Calvo benchmark (zero selection) decays exponentially at rate  $\bar{\Lambda}$ . State-dependent economies converge faster

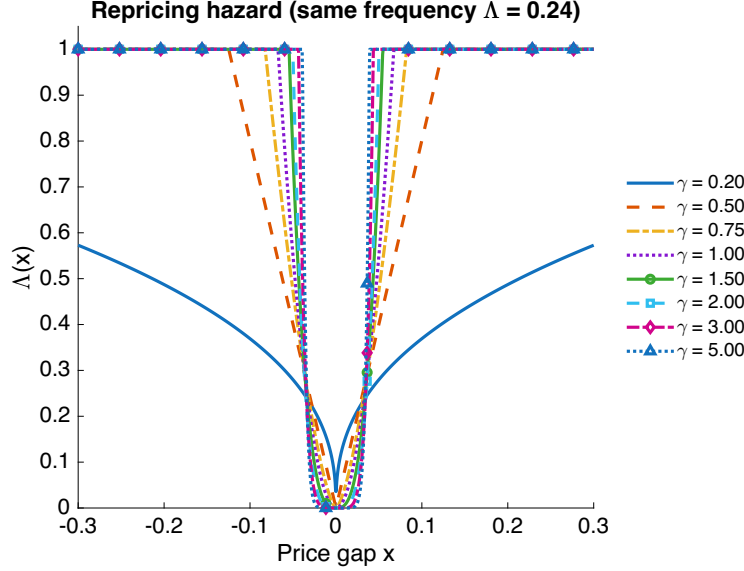


Figure 3: Repricing hazard across the spectrum

*Notes:* Repricing hazard  $\Lambda(x)$  for selected values of  $\gamma$ . All economies calibrated to  $\bar{\Lambda} = 0.24$  quarterly. Low  $\gamma$  produces a nearly flat hazard (close to Calvo); high  $\gamma$  produces a steep hazard (close to Golosov-Lucas).

because repricing is targeted at the most mispriced firms. The gap between Calvo and the state-dependent economy widens with  $\gamma$ : at the benchmark  $\hat{\gamma} = 1$  the output gap is approximately 37% of the Calvo path — selection is the dominant force; at the kurtosis-matching alternative  $\hat{\gamma} = 0.4$  it is approximately 75% of Calvo.

Figure D.2 in the Online Appendix shows the distribution of price changes for several values of  $\gamma$ ; the kurtosis-matching alternative  $\hat{\gamma} = 0.4$  produces a moderately leptokurtic distribution with raw kurtosis around 5, while the benchmark  $\hat{\gamma} = 1$  produces a less leptokurtic distribution closer to the heterogeneity-corrected kurtosis  $\approx 2$  that ALO recover from CPI microdata.

**Relation to Alvarez, Le Bihan, and Lippi (2016)** The Alvarez–Le Bihan–Lippi (ALL) sufficient statistic characterises non-neutrality in terms of the kurtosis and frequency of price changes. Within the power law family, both are endogenous functions of  $\gamma$  and  $\bar{\xi}$ . The present framework therefore provides the structural layer from which ALL’s observable diagnostics are derived:  $\gamma$  determines the kurtosis-frequency ratio, and hence the cumulative impulse response (CIR), through the shape of the repricing hazard. The practical value of  $\gamma$  over the ALL moments is that  $\gamma$  is a single structural parameter with a clear interpretation: the shape of the cost distribution. The ALL ratio combines two moments that can move independently in response to changes in the economic environment. For instance, changes in trend inflation



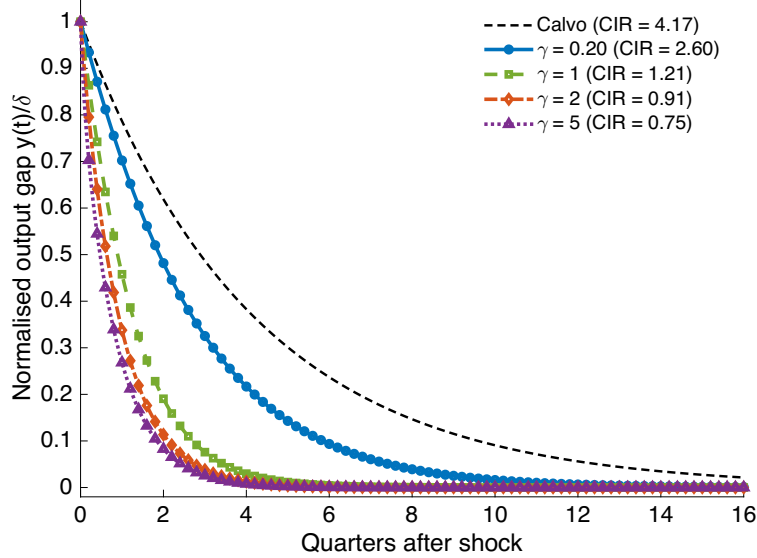


Figure 4: Output impulse response to a 1% monetary shock

*Notes:* All economies calibrated to  $\bar{\Lambda} = 0.24$  quarterly; they differ only in  $\gamma$ . Higher  $\gamma$  implies stronger selection, faster price-level convergence, and smaller cumulative output effects. CIR values computed via [Alvarez, Le Bihan, and Lippi \(2016\)](#):  $\text{CIR} = \text{Kurt}/(6 \cdot \text{Freq})$ . The Calvo benchmark (dashed) has zero selection.

shift both frequency and kurtosis, but the mapping between the two depends on  $G$ .

**Robustness on Dominick’s scanner data** A moment exercise on the Dominick’s Finer Foods scanner dataset ([Midrigan, 2011](#); [Chevalier, Kashyap, and Rossi, 2003](#)) (Online Appendix E) yields a non-sale kurtosis of 7.6 at a quarterly repricing frequency of 0.72 — substantially higher than CPI moments. Recalibrated to Dominick’s frequency, simulated kurtosis is essentially flat across  $\gamma \in [0.10, 0.40]$ : the heterogeneity-driven kurtosis variation flattens at high frequency, so kurtosis bounds  $\gamma \lesssim 0.4$  but does not pin it down further. The CPI/Dominick’s gap is consistent with sectoral heterogeneity in  $G$ , paralleling [Oskolkov and Lippi \(2026\)](#)’s plant-level investment evidence; modelling sector-specific  $(\gamma_s, \bar{\xi}_s)$  explicitly is a natural next step.

## 4.5 Global non-linear solution of the dynamic spectrum

This subsection executes the constructive payoff of the duality. The  $C^1$  regularity of the ex-ante value function (Theorem 2) and the closed-form smooth FOC  $S = C'(\varphi)$  of Proposition 1 together license a fully non-linear global solution of the dynamic model of Section 3.1 across the entire Calvo–Goloso–Lucas spectrum, by the repeated transition method (RTM) of [Lee \(2026\)](#). The previous subsections derived the closed-form Phillips-curve decomposition under

the static-surplus approximation  $S(x) = Bx^2$  and identified the benchmark  $\hat{\gamma} = 1$  from the hazard-exponent route of [Alvarez, Lippi, and Oskolkov \(2022\)](#). This subsection (i) reports the global solution at the benchmark calibration and positions it against existing approaches to solving the spectrum, (ii) uses it to verify that the analytical decomposition holds as a tight numerical approximation in the dynamic model, and (iii) uses it to produce state-dependent generalised impulse responses — a margin of behaviour that a linearised solution eliminates by construction.

A note on the model used here versus the model used in the calibration. The CPI moment exercise of Section 4.2 includes idiosyncratic price shocks ( $\sigma_z = 0.04$  with cross-product  $CV = 1$ ), which generate the kurtosis used by the kurtosis-matching alternative; the global solution below sets  $\sigma_z = 0$  and lets trend inflation alone keep the cross-section non-degenerate, both because the differentiability and decomposition results of this paper do not require idiosyncratic smoothing, and because the cleaner setting isolates the analytical content the global exercise is meant to check. The parameter  $\gamma$  is the same object in both: it is the shape of the cost distribution  $G$ , and the hazard  $\Lambda(x) = G(Bx^2/w) = (Bx^2/(w\bar{\xi}))^\gamma$  is identified from the gap-conditional probability of adjustment regardless of which mechanism keeps  $f_{ss}$  non-degenerate. The benchmark  $\hat{\gamma} = 1$  from the hazard-exponent route is the value plugged into the global solution.

The duality and the differentiability result (Theorem 2, Corollary 3) make the full dynamic model of Section 3.1 solvable by a global nonlinear method. I solve it by the repeated transition method (RTM) of [Lee \(2026\)](#). RTM matches periods along a simulated equilibrium path by a one-dimensional summary of the endogenous distributional state. No law of motion for the cross-section is parameterised, and the solution is not linearised around steady state.

The smooth FOC-envelope structure of Section 3.1 is implemented directly. Each period traces both the value  $V$  and its gradient  $V_x$ . The matched conditional expectations of  $V'$  and  $V'_x$  deliver the surplus  $S = V^a - V^n$  — hence the hazard  $\Lambda = G(S/w)$  — and the resetter's first-order condition, with  $V_x$  updated by the envelope  $V_x = (1 - \Lambda)V_x^n$ . No individual-level grid search for  $x'^*$  is required. The scalar summary that drives the matching is the lagged price-dispersion index  $\Delta_{t-1}$ ; the full routine and a within-exogenous-state monotonicity check of  $V(x; X)$  in  $\Delta_{t-1}$  are in Online Appendix G.

The smoothness on which the routine rests is precisely what Theorem 2 delivers. With continuous  $G$ , the ex-ante value is already smooth. What the duality contributes is a closed-form smooth FOC that lets the routine work in the FOC-envelope representation throughout, rather than tracking the threshold rule of the realised binary decision. The Golosov-Lucas (degenerate- $G$ ) limit is the only case in which the ex-ante value itself loses smoothness,

acquiring a corner at the inaction-band boundary; the routine treats this limit continuously as the high- $\gamma$  endpoint of the spectrum.

**Position relative to prior work** By “global” I mean four things together:

- (1) the dynamic model is solved at every  $\gamma$  across the spectrum, not just at endpoints;
- (2) the cross-sectional distribution of price gaps is tracked along its actual simulated path, not approximated by a forecasting rule for  $f_t$ ;
- (3) aggregate dynamics are produced without log-linearising around steady state, in response to the actual sequence of stochastic preference, TFP, and monetary shocks;
- (4) the kink in the value function at the inaction-band boundary is not smoothed numerically.

I am not aware of any existing solution of the Calvo–Goloso–Lucas spectrum that combines all four features. [Goloso and Lucas \(2007\)](#) solve the menu-cost endpoint at steady state, with impulse responses computed locally. [Alvarez, Le Bihan, and Lippi \(2016\)](#) summarise non-neutrality through a first-order-perturbation sufficient statistic. [Midrigan \(2011\)](#) and [Nakamura and Steinsson \(2010\)](#) solve calibrated menu-cost models numerically but at specific points of the spectrum and with linearised aggregate dynamics. The general-purpose heterogeneous-agent toolkits of [Reiter \(2009\)](#) and [Auclert, Rognlie, and Straub \(2020\)](#) could in principle be used, but they need a smooth first-order condition for the resetter’s choice, which the standard fixed-cost formulation does not provide directly — the realised threshold rule injects a kink into the conditional value function. Two ingredients make a fully nonlinear global solution available here:

- (i) the convex-cost representation of this paper, which delivers a smooth first-order condition for the resetter without threshold-specific machinery;
- (ii) the matching method of [Lee \(2026\)](#), which handles the joint state  $(x, f_t)$  without imposing a functional form on the evolution of the cross-section.

The Goloso–Lucas limit is included as the high- $\gamma$  endpoint of the same routine.

The global solution delivers two analyses: a steady-state consistency check against the analytical decomposition of Section 3, and generalised impulse responses (GIRFs) that capture state-dependent transmission.

**Consistency with the analytical decomposition** The static decomposition is exact under two conditions stated in Section 3: (i) the repricing surplus is quadratic in  $x$ ,  $S(x) = Bx^2$ , so the hazard takes the power-law form  $\Lambda = (Bx^2/(w\bar{\xi}))^\gamma$ ; (ii) the stationary gap distribution  $f_{ss}$  is supported in the interior of the inaction band, so  $\kappa = (1 + 2\gamma)\bar{\Lambda}$  holds without a boundary correction. In the dynamic model neither condition holds exactly — the dynamic surplus is only approximately quadratic, and the support of  $f_{ss}$  is the inaction band itself rather than its strict interior — so the analytical decomposition becomes a numerical approximation that the global solution lets me evaluate. The global exercise also makes a third, analytical, property useful: (iii) the ratio  $\kappa/\bar{\Lambda} = 1 + 2\gamma$  does not depend on the curvature  $B$  of  $S$  (Corollary 6), so any discrepancy between the static and dynamic  $B$  leaves  $\kappa/\bar{\Lambda}$  untouched. I check (i)–(ii) numerically and exploit (iii) across the spectrum  $\gamma$ , with  $\bar{\xi}(\gamma)$  recalibrated to hold the average repricing frequency at  $\bar{\Lambda} = 0.10$ .

(i) The dynamic surplus is approximately quadratic near the optimum, so the hazard retains its power-law form to that approximation. Its local curvature is, however, roughly four times the static  $\varepsilon/2$ . This reflects the discounted accumulation of flow losses over the firm’s expected mispricing duration: the static  $B$  misses the level of curvature, but not its shape. The closed-form  $\kappa = (1 + 2\gamma)\bar{\Lambda}$  in the dynamic model is therefore valid as a tight approximation, not as an exact identity.

(ii) Mass of  $f_{ss}$  at the  $\Lambda = 1$  boundary is below  $10^{-5}$  throughout the spectrum. The boundary-correction term in  $\kappa$  is therefore numerically negligible.

(iii) Both  $\kappa$  and  $\bar{\Lambda}$  scale with  $B$  through the hazard, so the larger dynamic curvature drops out of the ratio. The selection share  $2\gamma/(1 + 2\gamma)$  rises from 17% at  $\gamma = 0.1$  to 86% at  $\gamma = 3$ . At the benchmark  $\hat{\gamma} = 1$  it equals  $\frac{2}{3} \approx 67\%$ , the headline value of the calibration above, intact to the approximation error in (i).

The cross-sectional welfare cost of adjustment  $\int C(\Lambda(x)) f_{ss}(x) dx$  rises with  $\gamma$  and saturates: stronger selection buys lower dispersion at a higher resource cost. Figure 5 collects these objects.

**Generalised impulse responses: a comparative-static exercise** Because RTM produces the full non-linear equilibrium path, GIRFs follow by paired simulation in the sense of [Koop, Pesaran, and Potter \(1996\)](#): from each conditioning date, two economies are simulated forward sharing all future shock draws, differing only in the period-1 innovation, and the response is averaged over conditioning dates within a regime. Figure 6 reports the GIRF to a one-standard-deviation preference (demand) innovation at the benchmark  $\hat{\gamma} = 1$  and at the kurtosis-matching alternative  $\hat{\gamma} = 0.4$ , presented as a comparative-static exercise across the spectrum. Conditioning is on the price-dispersion index  $\Delta_t = \int (P_i/P)^{-\varepsilon} f_t(x) dx$  — a

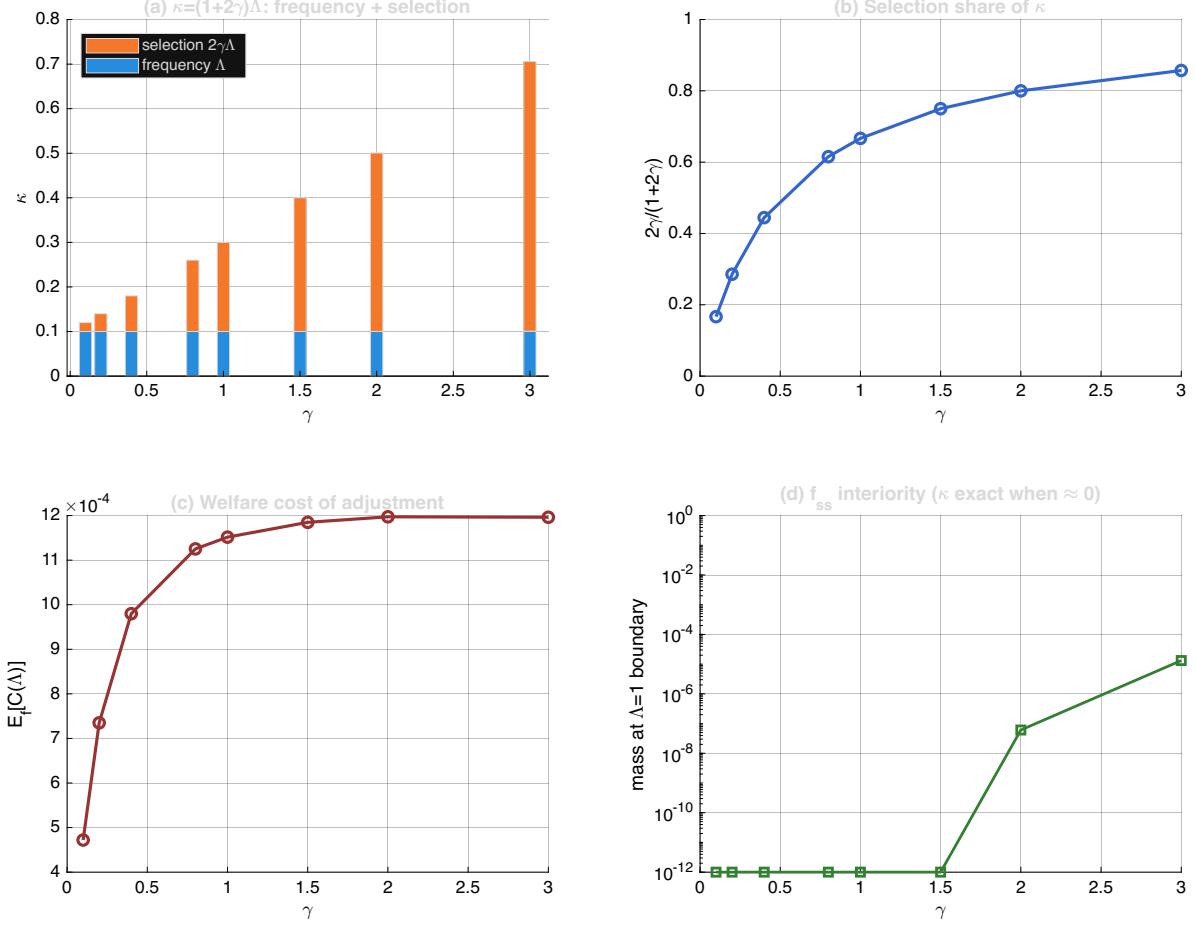


Figure 5: Steady-state spectrum comparative static across  $\gamma$

*Notes:* Steady-state objects across the spectrum index  $\gamma$ , with  $\bar{\xi}(\gamma)$  recalibrated to hold the average repricing frequency at  $\bar{\Lambda} = 0.10$ . (a) the Phillips slope  $\kappa = (1 + 2\gamma)\bar{\Lambda}$  decomposed into the frequency component  $\bar{\Lambda}$  and the selection wedge  $2\gamma\bar{\Lambda}$ ; (b) the selection share  $2\gamma/(1 + 2\gamma)$  ( $\frac{2}{3} \approx 67\%$  at the benchmark  $\gamma = 1$ ); (c) the welfare cost of adjustment  $\int C(\Lambda(x)) f_{ss}(x) dx$ ; (d) the mass of  $f_{ss}$  at the inaction-band boundary, negligible throughout, so the boundary correction to  $\kappa = (1 + 2\gamma)\bar{\Lambda}$  is numerically negligible.

moment of the endogenous cross-section  $f_t$ , and the same scalar summary the RTM matching uses. Base dates are split, within each TFP state, into the top and bottom quartile of  $\Delta_t$  (the “high- $\Delta$ ” and “low- $\Delta$ ” regimes). The demand shock raises both output and inflation on impact at both calibrations, with the selection margin  $\bar{\Lambda}$  rising as more firms cross the adjustment threshold.

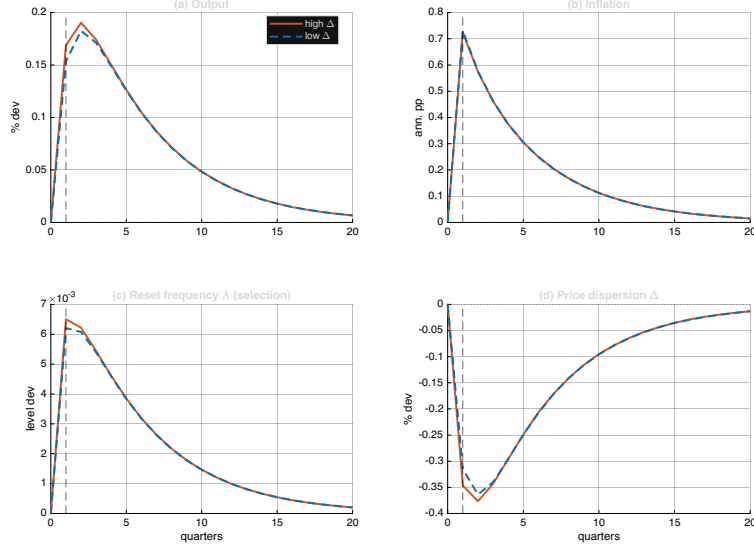
The state-dependence of transmission is *modest at both calibrations and weakens monotonically as  $\gamma$  rises*. At  $\hat{\gamma} = 0.4$  (selection share 44%), the high- $\Delta$  regime amplifies the impact output response by approximately 10% relative to the low- $\Delta$  regime (+0.17% vs +0.15%), with comparable amplification on real marginal cost and on price dispersion (both

$\approx 10\%$ ); the selection-margin gap is  $\approx 5\%$ , and the inflation response is essentially flat across regimes ( $-1.2\%$  relative gap). At the benchmark  $\hat{\gamma} = 1$  (selection share  $\frac{2}{3}$ ), the output gap shrinks to  $\approx 6\%$  and the selection-margin gap to  $\approx 1\%$ . An intermediate calibration at  $\hat{\gamma} = 0.5$  lies between the two ( $\approx 9\%$  output gap,  $\approx 4.5\%$  selection-margin gap), confirming the monotone trend. The economic interpretation is structural: as selection becomes the dominant flexibility margin, adjustment is concentrated on the most-mispriced firms regardless of the cross-section state, so the composition of the cross-section has less leverage over which firms a marginal shock tips into adjustment. This is consistent with the closed-form decomposition  $2\gamma/(1+2\gamma)$ , which depends only on  $\gamma$  and not on the cross-section state. The dynamic exercise confirms a small but non-zero state-dependence channel that a linearised solution eliminates by construction, with magnitude declining as the spectrum moves toward Golosov-Lucas.

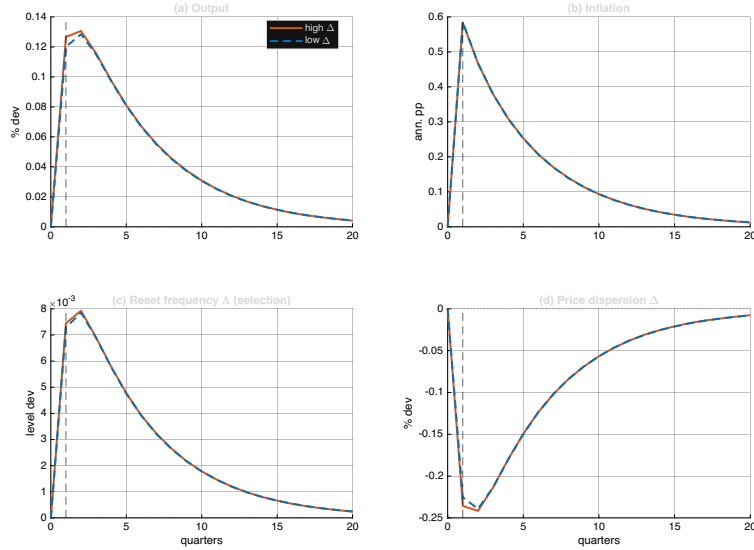
## 5 Discussion

**Further applications** Beyond pricing (and investment, in Online Appendix B), the duality applies to any extensive-margin decision with heterogeneous costs. Heterogeneous entry costs micro-found smooth aggregate entry: the free-entry condition  $S = wG^{-1}(\varphi)$  equates the marginal entrant’s cost to the marginal cost of raising the entry rate. Heterogeneous vacancy-creation costs micro-found convex posting costs (Pissarides, 2000), with  $C''$  reflecting the cost distribution rather than a technology. Heterogeneous participation costs generate smooth labour supply without Rogerson (1988) lotteries, with elasticity  $g(G^{-1}(\varphi))/w$ . The same logic extends to technology adoption and credit default (Bernanke, Gertler, and Gilchrist, 1999).

**Friction-extended duality and matching applications** When an aggregate friction  $F$  enters the operative surplus per attempted action, the dual cost and distribution rescale as  $\tilde{G}(z) = G(Fz)$  and  $\tilde{C}(\varphi) = C(\varphi)/F$ , and the decision-maker’s first-order condition becomes  $F S = C'(\varphi)$ . The search-and-matching environment is a canonical instantiation, with  $F = F(\theta)$  the matching probability and  $\theta$  the market tightness (Pissarides, 2000). The dual machinery absorbs the general-equilibrium friction without re-derivation: the random-cost and convex-cost representations of Alvarez, Lippi, and Oskolkov (2022) carry over verbatim once  $G$  and  $C$  are rescaled by  $F$ . When  $F$  responds to the aggregate adjustment rate through  $\theta$ , the planner’s first-order condition adds a congestion-externality term  $(dF/d\varphi) S \varphi$ , and the Pigouvian entry tax  $\tau^*(\varphi) = -(dF/d\varphi) S \varphi$  decentralises the planner. Green/non-green technology adoption is another canonical instantiation: when aggregate adoption shifts  $F$



(a) Kurtosis-matching alternative  $\hat{\gamma} = 0.4$  (selection share 44%)



(b) Benchmark calibration  $\hat{\gamma} = 1$  (selection share  $\frac{2}{3}$ )

Figure 6: State-dependent GIRFs to a preference (demand) shock: comparative static across the spectrum

*Notes:* Generalised impulse responses to a +1-standard-deviation preference innovation, computed by paired Monte-Carlo simulation along the converged RTM path, percentile-averaged over conditioning dates and split by the price-dispersion regime (high  $\Delta$  = top quartile of  $\Delta_t$  within TFP; low  $\Delta$  = bottom quartile). Horizon 0 is the pre-shock baseline (identically zero in the paired construction); the thin dashed vertical marks the shock impact at  $j = 1$ . Panels within each subfigure: (a) output, (b) inflation, (c) reset frequency  $\bar{\Lambda}$  (the selection margin), (d) price dispersion  $\Delta$ . State-dependence is modest at both calibrations and weakens as  $\gamma$  rises, consistent with selection becoming the dominant flexibility margin at higher  $\gamma$ ; an intermediate  $\hat{\gamma} = 0.5$  run (not shown) interpolates the trend on every variable.

through emissions spillovers, learning-by-doing, or scale economies in green infrastructure, the same wedge appears as an emissions tax or green-adoption subsidy (Acemoglu et al., 2012). With Cobb–Douglas matching the wedge collapses to the Hosios (1990) efficiency condition, and the non-Cobb–Douglas extension of Mangin and Julien (2021) follows from the same algebra. The efficiency results recovered here are known; the contribution is representational—the dual construction reaches general-equilibrium matching frictions without re-derivation (see Online Appendix H).

**Structural discipline and identifiability** The  $G$  framing makes distributional assumptions transparent: quadratic costs require uniform  $G$ ; iso-elastic costs require power law  $G$ . It supplies a common language — the Calvo–Golosov-Lucas debate, selection in non-neutrality, and the Khan and Thomas (2008) irrelevance result are all statements about  $G$  — and an identifiable primitive:  $G$  is fully recovered from the adjustment hazard (Section 4.1) and disciplined by price-change size distributions and investment spike frequencies (model-dependent). Oskolkov and Lippi (2026) recover  $G$  from plant-level data, where the duality implies the same  $G$  simultaneously determines  $C$ , the investment hazard, and capital misallocation. The aggregate smoothness emphasised by Auclert, Rognlie, and Straub (2020) has an individual-level structural counterpart here: whenever  $G$  has a positive density, each agent’s problem is smooth even though realised actions remain binary; the cross-sectional aggregation step is separate.

**Scope and limitations** The equivalence is an identity of value functions and adjustment probabilities, not of welfare, interpretation, or causality. The identifiable object is  $G$  (equivalently  $C$ ), not the “type” of friction. Random-menu-cost and probability-choice economies make different predictions about how  $G$  responds to changes in the environment; endogenising  $G$  is a natural next step. The global solution of Section 4.5 keeps the cross-section non-degenerate by trend inflation alone; adding idiosyncratic price shocks would change the calibrated curvature  $B$  but, by the curvature-invariance of  $\kappa/\bar{\Lambda} = 1 + 2\gamma$ , would not change the frequency–selection decomposition itself. The novelty claim of ‘sec:global’ — a global solution of the spectrum including the Golosov-Lucas limit — is stated relative to the existing literature surveyed there; if closer prior work exists that I have missed, the positioning paragraph in Section 4.5 should be read with that caveat.



## 6 Conclusion

The equivalence established in this paper changes how one should think about adjustment frictions. The distinction between fixed and convex costs is not a distinction between two kinds of economic friction. It is a distinction between two representations of the same friction, linked ex-ante by Fenchel conjugacy. The fixed cost distribution  $G$  is the structural primitive that governs both.

Three implications follow. First, the lumpy–smooth divide dissolves at the ex-ante extensive margin. Realised actions remain binary at the micro level; what becomes smooth and convex is the ex-ante extensive margin, whenever fixed costs are absolutely continuous with positive density. The deterministic fixed cost — the textbook case — is the singular exception. This recasts the standard view that convex costs approximate the “true” lumpy friction: at the ex-ante extensive margin, the deterministic fixed cost is the knife-edge limit and arbitrarily close perturbations restore strict convexity. The same observation provides a partial-equilibrium foundation for the irrelevance result of [Khan and Thomas \(2008\)](#): with uniform fixed costs, each firm’s ex-ante extensive margin is already convex, so aggregation is not the source of smoothness.

Second, the same mapping  $G \mapsto C$  shows that the cost specification in each of menu costs, lumpy investment, discrete choice, rational inattention, and control costs is an implicit distributional assumption on  $G$  — one that can be made explicit, tested, and recovered from cost-distribution data. Under the fixed-cost primitive,  $C(\varphi^*)$  acquires a concrete resource-cost interpretation, and the smooth first-order condition  $S = C'(\varphi)$  replaces the threshold-tracking machinery that standard menu-cost implementations require.

Third, in the pricing application under the static-surplus approximation, the equivalence gives the repricing hazard in closed form and enables an analytical decomposition of non-neutrality into frequency and selection. This decomposition rests on the explicit functional form  $\Lambda(x) = G(Bx^2/w)$ .

This paper opens a research agenda along several directions that the duality and the global solution put in reach. One is to endogenise  $G$  as a choice of menu-cost technology that responds to the inflation environment, with possible connections to the Lucas critique and to [Vavra \(2014\)](#)’s evidence on the co-movement of price-change dispersion and inflation. Another is sector-level identification of  $\gamma_s$  from sector-specific micro moments, motivated by the gap between the Dominick’s scanner moments and the CPI moments documented above, which could help shed light on whether the benchmark  $\hat{\gamma} = 1$  (or the kurtosis-matching alternative  $\hat{\gamma} \approx 0.4$ ) is a population parameter or an aggregation of heterogeneous sectoral values.

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